

Sid Mathur's Thematic ETF Screener – Daily Report

2026-06-19

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Action Center

Run date: 2026-06-19 **Universe:** 660 ETFs / 26 clusters

Full PC1: FLAT **Clusters long:** 9/26 **ETFs long:** 257/660 (39%)

Today's events (cluster level)

spread_buy = 5dma – 50dma; spread_sell = 5dma – 20dma (price units). A BUY fired today iff spread_buy crossed above 0; a SELL iff spread_sell crossed below the cushioned threshold.

SELL (1)

cluster	theme	spread_sell	p_h5	nss
cluster_26	Leveraged / Inverse	-0.284	0.72	-0.718

Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p_h5 is the 5-day probability for context. **spread_buy = 5dma – 50dma, spread_sell = 5dma – 20dma** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

Imminent BUY (top 5 by h=1 NSS, filter-approved)

cluster	theme	spread_buy	p_h1	p_h5	nss_h1	nss_h5
cluster_14	Financials	4.120	1	1.00	1.000	1.00
cluster_1	U.S. Mid Cap Equity	5.228	1	1.00	1.000	1.00
cluster_21	Japan Equity	7.316	1	1.00	1.000	0.96
cluster_3	Developed ex-US Equity	2.268	1	0.94	0.957	0.73
cluster_15	Real Estate	1.648	1	0.91	0.952	0.74

Imminent SELL (top 3 by h=1 NSS)

cluster	theme	spread_sell	p_h1	p_h5	nss_h1	nss_h5
cluster_22	Broad Commodities	-10.416	1.000	1.00	-1.000	-1.00
cluster_25	International Fixed Income	-0.289	0.999	0.85	-0.999	-0.85
cluster_9	Asia-Pacific Equity	-0.149	0.622	0.45	-0.622	-0.43

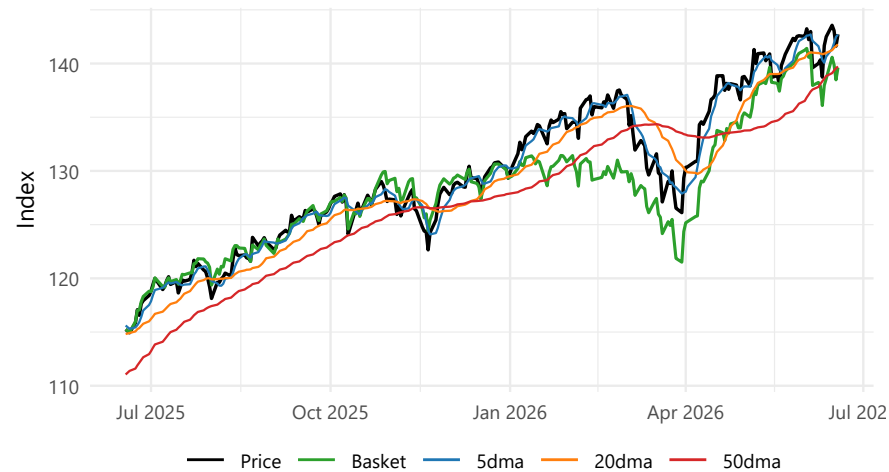
Full Sample PC1

Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.94 sticky

Current state: FLAT

5dma: 142.691 20dma: 141.777 50dma: 139.728

Spread vs 50dma: +2.962 Spread vs 20dma: +0.914

P(buy crossover within 5 days): 99.2% P(sell): 10.3%

Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	pos_wt	cor
VHT	cluster_12	Healthcare	0.999	30	2.9%	8.79%	0.93
VTV	cluster_2	U.S. Large Cap Value	0.997	47	3.6%	14.36%	0.98
MUB	cluster_16	Municipal Bond	0.979	15	0.4%	18.09%	0.95
BND	cluster_4	Aggregate Fixed Income	0.849	1	0.3%	20.14%	0.99
XLP	cluster_19	Consumer Staples	0.557	3	-1.5%	6.90%	0.99
XLU	cluster_19	Consumer Staples	0.557	3	-1.5%	4.27%	0.99
HYG	cluster_11	High Yield Bond	0.542	1	0.3%	18.33%	0.95
BBAX	cluster_9	Asia-Pacific Equity	-0.431	18	-1.6%	4.43%	0.97
IGF	cluster_9	Asia-Pacific Equity	-0.431	18	-1.6%	2.83%	0.97
EWZ	cluster_9	Asia-Pacific Equity	-0.431	18	-1.6%	0.93%	0.97
BWX	cluster_25	International Fixed Income	-0.852	14	-2.0%	15.02%	0.93
PDBC	cluster_22	Broad Commodities	-1.000	10	-9.7%	5.60%	0.98

Watchlist (imminent-buy clusters, $p > 80\%$, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
XLF	cluster_14	Financials	1.00	1.000	1.097	0.96
QQQ	cluster_6	U.S. Large Cap Growth	1.00	0.521	1.022	0.98
IJH	cluster_1	U.S. Mid Cap Equity	1.00	1.000	1.007	0.99
EWJ	cluster_21	Japan Equity	1.00	0.956	0.909	0.97
IEMG	cluster_8	Emerging Markets Equity	0.96	0.774	0.924	0.99
VEA	cluster_3	Developed ex-US Equity	0.94	0.730	0.966	0.99
VNQ	cluster_15	Real Estate	0.91	0.739	0.971	0.99
VXF	cluster_7	U.S. Extended Market Equity	0.88	0.212	1.070	0.96
VOO	cluster_5	U.S. Large Cap Equity	0.88	0.247	0.958	0.99

Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
1	cluster_1	U.S. Mid Cap Equity	U.S. Mid Cap Equi...	92	FLAT				1.000	-0.000
14	cluster_14	Financials	Financials	14	FLAT				1.000	+0.000
12	cluster_12	Healthcare	Healthcare	17	LONG		30	2.9%	0.999	-0.001
2	cluster_2	U.S. Large Cap Value	U.S. Large Cap Va...	73	LONG		47	3.6%	0.997	-0.003
16	cluster_16	Municipal Bond	Municipal Bond	13	LONG		15	0.4%	0.979	+0.013
21	cluster_21	Japan Equity	Japan Equity	9	FLAT				0.956	+0.171
4	cluster_4	Aggregate Fixed Income	Aggregate Fixed I...	58	LONG		1	0.3%	0.849	+0.045
8	cluster_8	Emerging Markets Equity	Emerging Markets ...	25	FLAT				0.774	+0.178
15	cluster_15	Real Estate	Real Estate	14	FLAT				0.739	-0.199
3	cluster_3	Developed ex-US Equity	Developed ex-US E...	61	FLAT				0.730	-0.030
19	cluster_19	Consumer Staples	Consumer Staples	11	LONG		3	-1.5%	0.557	-0.120
11	cluster_11	High Yield Bond	High Yield Bond	21	LONG		1	0.3%	0.542	+0.181
6	cluster_6	U.S. Large Cap Growth	U.S. Large Cap Gr...	39	FLAT				0.521	+0.166
24	cluster_24	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	9	FLAT				0.443	+0.128
5	cluster_5	U.S. Large Cap Equity	U.S. Large Cap Eq...	57	FLAT				0.247	+0.165
7	cluster_7	U.S. Extended Market Equity	U.S. Extended Mar...	28	FLAT				0.212	+0.133
18	cluster_18	India Equity	India Equity	12	FLAT				-0.184	+0.240
9	cluster_9	Asia-Pacific Equity	Asia-Pacific Equi...	23	LONG		18	-1.6%	-0.431	-0.139
23	cluster_23	Bank Loan / Floating Rate	Bank Loan / Float...	9	FLAT				-0.538	+0.072
10	cluster_10	Materials	Materials	21	FLAT				-0.715	-0.008
26	cluster_26	Leveraged / Inverse	Leveraged / Inver...	2	FLAT				-0.718	-0.193
25	cluster_25	International Fixed Income	International Fix...	6	LONG		14	-2.0%	-0.852	-0.184
17	cluster_17	Gold	Gold	12	FLAT				-0.863	+0.090
20	cluster_20	Inflation-Protected Fixed I...	Inflation-Protect...	10	FLAT				-0.938	+0.009
13	cluster_13	China Equity	China Equity	15	FLAT				-0.946	+0.030
22	cluster_22	Broad Commodities	Broad Commodities	9	LONG		10	-9.7%	-1.000	-0.000

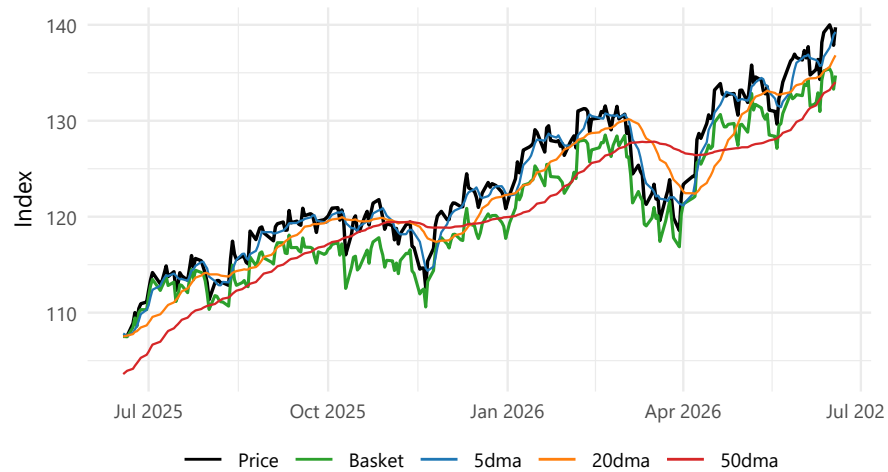
Cluster Details

Cluster 1 – U.S. Mid Cap Equity

Members: 92 **Var explained:** 0.867 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +41.7%)
- Latest sell_cross: 2026-05-19
- Current 252d return: +30.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-19 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Industrials	15	0.1534
U.S. Small Cap Equity	10	0.1140
U.S. Mid Cap Equity	9	0.1032
U.S. Small Cap Value	6	0.0673
U.S. Mid Cap Growth	6	0.0660
Consumer Discretionary	6	0.0602
U.S. Small Cap Growth	3	0.0343
U.S. Mid Cap Value	3	0.0341

U.S. Momentum Equity	3	0.0294
Leveraged / Inverse	2	0.0216
U.S. Small Cap Momentum Equity	2	0.0215
Momentum Equity	2	0.0205
Unknown	1	0.0116
U.S. Small Cap ESG Equity	1	0.0116
U.S. Mid/Small Cap Equity	1	0.0115
U.S. Small Cap Fundamental Equity	1	0.0115
U.S. Small Cap Multifactor Equity	1	0.0115
U.S. Small/Mid Cap Fundamental Equity	1	0.0115
Mid Cap Value Equities	1	0.0115
U.S. ESG Equity	1	0.0113
Dividend Equity	1	0.0112
U.S. Size Factor Equity	1	0.0112
U.S. Mid Cap Quality Equity	1	0.0112
U.S. Mid Cap Dividend Equity	1	0.0111
U.S. Small/Mid Cap Equity	1	0.0111
Small Cap Value Equities	1	0.0111
U.S. Equity	1	0.0110
U.S. Small Cap Industrials	1	0.0109
Value Equity	1	0.0108
U.S. Small Cap Dividend Equity	1	0.0108
U.S. Mid Cap Momentum Equity	1	0.0107
U.S. Micro Cap Equity	1	0.0106
U.S. Moat Equity	1	0.0103
U.S. Small Cap Consumer Discretionary	1	0.0103
Technology	1	0.0103
Growth Equity	1	0.0101
IPO Equity	1	0.0098

Replication basket – IJH (cor=0.99) basket lead at lag 1: -0.007

ticker	rep_wt	name	category	etf_signal
IJH	1.007	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
24	SCHM	0.0116	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
30	NUSC	0.0116	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	FLAT
33	MDY	0.0116	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	FLAT

40	IVOO	0.0116	Vanguard S&P Mid-Cap 400 ETF		FLAT
44	IJH	0.0116	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
45	SPMD	0.0116	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	FLAT
47	VB	0.0116	Vanguard Small-Cap ETF	U.S. Small Cap Equity	FLAT
18	JHMM	0.0115	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	FLAT
26	FNX	0.0115	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	FLAT
31	PRFZ	0.0115	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	FLAT
32	SMLF	0.0115	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	FLAT
48	SCHA	0.0115	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
60	EZM	0.0115	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	FLAT
65	FNDA	0.0115	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	FLAT
74	VBR	0.0115	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	FLAT
4	IWM	0.0114	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	FLAT
19	IWR	0.0114	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	LONG
34	IJT	0.0114	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	FLAT
35	IJK	0.0114	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
36	VIOG	0.0114	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	FLAT
38	MDYG	0.0114	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
39	SLYG	0.0114	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	FLAT
41	VIOO	0.0114	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	FLAT
42	VTWO	0.0114	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	FLAT
43	IJR	0.0114	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT
46	SPSM	0.0114	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	FLAT
67	IJJ	0.0114	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
69	IVOV	0.0114	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
70	MDYV	0.0114	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	FLAT
23	IMCB	0.0113	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	LONG
27	FYX	0.0113	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	FLAT
37	IVOG	0.0113	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
61	PSC	0.0113	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
66	IWN	0.0113	iShares Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
76	ETHO	0.0113	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	FLAT
84	RWK	0.0113	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	FLAT
15	FXR	0.0112	First Trust Industrials AlphaDEX	Industrials	LONG
20	SIZE	0.0112	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	LONG
25	VO	0.0112	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	LONG
73	VTWV	0.0112	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
80	XMHQ	0.0112	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	FLAT
91	SDVY	0.0112	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	LONG
58	DON	0.0111	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	FLAT
59	EES	0.0111	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	FLAT
68	IJS	0.0111	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
71	SLYV	0.0111	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	FLAT
72	VIOV	0.0111	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	FLAT
77	RWJ	0.0111	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	FLAT

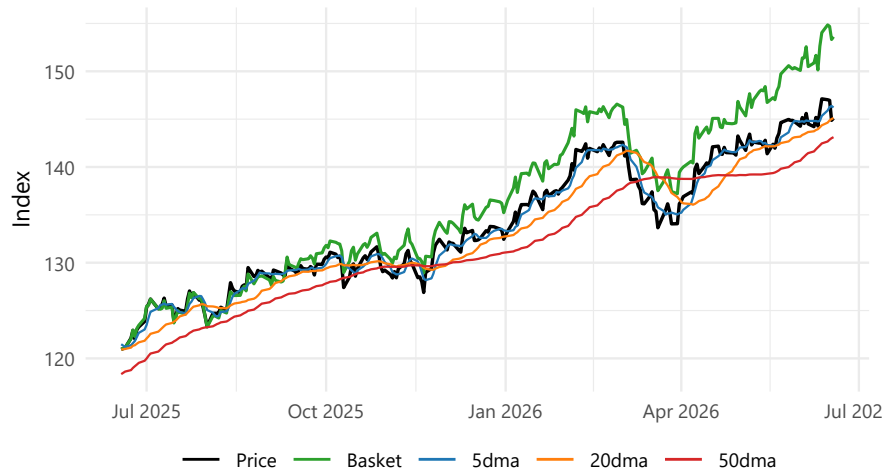
92	ISMD	0.0111	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
17	IYJ	0.0110	iShares U.S. Industrials ETF	Industrials	FLAT
21	VIS	0.0110	Vanguard Industrials ETF	Industrials	FLAT
22	FIDU	0.0110	Fidelity MSCI Industrials Index ETF	Industrials	FLAT
85	OMFS	0.0110	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	FLAT
14	IMCG	0.0109	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
49	FXD	0.0109	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	LONG
78	PSCI	0.0109	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	FLAT
79	PAVE	0.0109	Global X U.S. Infrastructure Developmen...	Industrials	FLAT
83	RFV	0.0109	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	FLAT
89	XSMO	0.0109	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	FLAT
57	DES	0.0108	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	FLAT
90	FYT	0.0108	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	FLAT
1	XLI	0.0107	State Street Industrial Select Sector S...	Industrials	FLAT
10	XMMO	0.0107	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	LONG
56	RSPD	0.0107	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	LONG
28	DWAS	0.0106	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	FLAT
29	IWC	0.0106	iShares Microcap ETF	U.S. Micro Cap Equity	FLAT
81	VFMO	0.0106	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	LONG
86	RZV	0.0106	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	FLAT
11	IWP	0.0105	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG
13	VOT	0.0105	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
51	FTXR	0.0105	First Trust Nasdaq Transportation ETF	Industrials	LONG
5	FV	0.0103	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
6	FTC	0.0103	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
16	MOAT	0.0103	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	FLAT
82	PKB	0.0103	Invesco Building & Construction ETF	Industrials	LONG
88	PSCD	0.0103	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	LONG
7	PDP	0.0102	Invesco Dorsey Wright Momentum ETF	Momentum Equity	LONG
64	XTN	0.0102	State Street SPDR S&P Transportation ETF	Industrials	FLAT
9	RPG	0.0101	Invesco S&P 500 Pure Growth ETF	Growth Equity	LONG
87	PRN	0.0101	Invesco Dorsey Wright Industrials Momen...	Industrials	FLAT
2	XRT	0.0100	State Street SPDR S&P Retail ETF	Consumer Discretionary	LONG
50	PEJ	0.0100	Invesco Leisure and Entertainment ETF	Consumer Discretionary	LONG
55	IYT	0.0100	iShares U.S. Transportation ETF	Industrials	FLAT
8	FPX	0.0098	First Trust US Equity Opportunities ETF	IPO Equity	FLAT
63	XHB	0.0098	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	LONG
75	QMOM	0.0095	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	FLAT
52	PPA	0.0093	Invesco Aerospace & Defense ETF	Industrials	LONG
62	XAR	0.0093	State Street SPDR S&P Aerospace & Defen...	Industrials	LONG
12	MTUM	0.0092	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	LONG
3	JETS	0.0091	U.S. Global Jets ETF	Industrials	FLAT
54	ITB	0.0089	iShares U.S. Home Construction ETF	Consumer Discretionary	LONG
53	ITA	0.0087	iShares U.S. Aerospace & Defense ETF	Industrials	LONG

Cluster 2 – U.S. Large Cap Value

Members: 73 **Var explained:** 0.874 **Signal:** LONG since 2026-04-13 (47 days, 3.6%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +22.3%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +20.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)

**Category breakdown**

category	n_members	abs_weight
U.S. Dividend Equity	22	0.3011
U.S. Large Cap Value	10	0.1439
Value Equity	4	0.0557
U.S. Low Volatility Equity	4	0.0518
U.S. Multifactor Equity	3	0.0421
U.S. Value Factor Equity	3	0.0413
Dividend Equity	3	0.0401
Utilities	3	0.0383
U.S. Fundamental Equity	2	0.0287
U.S. Mid Cap Value	2	0.0286
U.S. Equity	2	0.0285

U.S. Equal Weight Equity	2	0.0284
U.S. Small Cap Low Volatility	2	0.0261
U.S. Mega Cap Value	1	0.0142
U.S. Revenue Weighted Equity	1	0.0141
Large Cap Blend Equities	1	0.0140
U.S. Buyback Equity	1	0.0138
U.S. Large Cap Equity	1	0.0137
U.S. Quality / Factor Equity	1	0.0135
U.S. Mid Cap Low Volatility	1	0.0135
U.S. Defensive Equity	1	0.0132
Global Equity	1	0.0127
Communication Services	1	0.0115
Real Estate	1	0.0111

Replication basket – VTV (cor=0.98) basket lead at lag 1: 0.006

ticker	rep_wt	name	category	etf_signal
VTV	0.997	Vanguard Value ETF	U.S. Large Cap Value	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
29	IWD	0.0145	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
31	JPUS	0.0145	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	FLAT
35	VONV	0.0145	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
41	SCHV	0.0145	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG
3	RSP	0.0144	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
14	PRF	0.0144	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
21	NULV	0.0144	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	FLAT
30	IVE	0.0144	iShares S&P 500 Value ETF	U.S. Large Cap Value	LONG
39	ILCV	0.0144	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	LONG
40	IUSV	0.0144	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	LONG
43	VTV	0.0144	Vanguard Value ETF	U.S. Large Cap Value	LONG
59	VOE	0.0144	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
64	DTD	0.0144	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
69	DGRO	0.0144	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
19	CFA	0.0143	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	FLAT
25	FNDX	0.0143	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
28	ONEV	0.0143	State Street SPDR Russell 1000 Low Vola...	U.S. Equity	FLAT
33	VOOV	0.0143	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	LONG
38	VYM	0.0143	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	LONG

42	SPYV	0.0143	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	LONG
63	DLN	0.0143	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
17	AIVL	0.0142	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
27	IWX	0.0142	iShares Russell Top 200 Value ETF	Value Equity	LONG
36	MGV	0.0142	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	LONG
44	FTA	0.0142	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	LONG
55	IWS	0.0142	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
56	ONEY	0.0142	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	LONG
72	JPME	0.0142	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	FLAT
15	RWL	0.0141	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
61	PFM	0.0141	Invesco Dividend Achievers ETF	U.S. Dividend Equity	LONG
6	FEX	0.0140	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
22	SDY	0.0140	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
26	EQAL	0.0140	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	FLAT
37	SCHD	0.0140	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
4	FVD	0.0139	First Trust VL Dividend	U.S. Dividend Equity	LONG
23	FDVV	0.0139	Fidelity High Dividend ETF	U.S. Dividend Equity	LONG
47	SDOG	0.0139	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	LONG
49	DVY	0.0139	iShares Select Dividend ETF	U.S. Dividend Equity	FLAT
70	VIG	0.0139	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
5	PKW	0.0138	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG
10	PWV	0.0138	Invesco Large Cap Value ETF	Value Equity	LONG
73	JVAL	0.0138	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
66	DIA	0.0137	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	LONG
11	RDVY	0.0136	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	FLAT
13	REGL	0.0136	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	LONG
58	SPYD	0.0136	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	LONG
68	LGLV	0.0136	State Street SPDR US Large Cap Low Vola...	U.S. Low Volatility Equi...	LONG
51	RPV	0.0135	Invesco S&P 500 Pure Value ETF	Value Equity	LONG
53	XMLV	0.0135	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	LONG
60	FTCS	0.0135	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	LONG
62	OUSA	0.0135	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
16	DHS	0.0134	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	FLAT
20	CFO	0.0134	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	FLAT
57	SMMV	0.0134	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
32	VLUE	0.0133	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
18	CDC	0.0132	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	FLAT
50	RDIV	0.0132	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	LONG
67	USMV	0.0132	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	FLAT
52	SPHD	0.0131	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
9	FIW	0.0130	First Trust Water ETF	Utilities	LONG
12	FDL	0.0130	First Trust Morningstar ETF	U.S. Dividend Equity	LONG
45	PEY	0.0130	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	FLAT
7	PHO	0.0129	Invesco Water Resources ETF	Utilities	FLAT
46	DIV	0.0129	Global X Super Dividend ETF	U.S. Dividend Equity	FLAT

2	ACWV	0.0127	iShares MSCI Global Min Vol Factor ETF	Global Equity	FLAT
24	LVHD	0.0127	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	LONG
34	HDV	0.0127	iShares Core High Dividend ETF	U.S. Dividend Equity	FLAT
54	XSLV	0.0127	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	FLAT
48	SMDV	0.0126	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT
8	CGW	0.0123	Invesco S&P Global Water Index ETF	Utilities	FLAT
65	SPLV	0.0123	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	LONG
71	IYZ	0.0115	iShares U.S. Telecommunications ETF	Communication Services	FLAT
1	REM	0.0111	iShares Mortgage Real Estate ETF	Real Estate	FLAT

Cluster 3 – Developed ex-US Equity

Members: 61 **Var explained:** 0.896 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +47.7%)
- Latest sell_cross: 2026-06-10
- Current 252d return: +25.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-10 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
International Equity	19	0.3158
Developed ex-US Equity	9	0.1528
Global ex-US Equity	5	0.0848
Europe Equity	4	0.0675
International Dividend Equity	4	0.0659
Alternatives	4	0.0615
Eurozone Equity	3	0.0490
Commodity	1	0.0172
Global ex-US Small Cap Equity	1	0.0166
Germany Equity	1	0.0162
France Equity	1	0.0162
Italy Equity	1	0.0159
United Kingdom Equity	1	0.0158
Europe Financials	1	0.0158
Netherlands Equity	1	0.0157
Sweden Equity	1	0.0156
Spain Equity	1	0.0150
Austria Equity	1	0.0148
Belgium Equity	1	0.0146
Poland Equity	1	0.0135

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.017

ticker	rep_wt	name	category	etf_signal
VEA	0.966	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	EFA	0.0172	iShares MSCI EAFE ETF	Developed ex-US Equity	FLAT
3	VEA	0.0172	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT
40	GSIE	0.0172	Goldman Sachs ActiveBeta International ...	Commodity	LONG
41	ESGD	0.0172	iShares ESG Aware MSCI EAFE ETF	International Equity	FLAT
51	IEFA	0.0172	iShares Core MSCI EAFE ETF	Developed ex-US Equity	FLAT
52	SCHF	0.0172	Schwab International Equity ETF	Developed ex-US Equity	FLAT
53	SPDW	0.0172	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	FLAT
6	IEV	0.0170	iShares Europe ETF	Europe Equity	FLAT
16	DWM	0.0170	WisdomTree International Equity Fund	International Dividend E...	LONG

19	PXF	0.0170	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	FLAT
25	TLTD	0.0170	FlexShares Morningstar Developed Market...	International Equity	LONG
29	ACWX	0.0170	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	FLAT
32	INTF	0.0170	iShares International Equity Factor ETF	International Equity	LONG
46	IEUR	0.0170	iShares Core MSCI Europe ETF	Europe Equity	FLAT
47	IXUS	0.0170	iShares Core MSCI Total International S...	Global ex-US Equity	FLAT
48	VEU	0.0170	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	FLAT
49	VGK	0.0170	Vanguard FTSEEuropean ETF	Europe Equity	FLAT
50	VXUS	0.0170	Vanguard Total International Stock ETF	Global ex-US Equity	FLAT
30	CWI	0.0169	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	FLAT
33	IQLT	0.0169	iShares MSCI Intl Quality Factor ETF	International Equity	LONG
35	QEFA	0.0169	State Street SPDR MSCI EAFE StrategicFa...	International Equity	FLAT
39	FNDF	0.0169	Schwab Fundamental International Equity...	International Equity	FLAT
14	EZU	0.0168	iShares MSCI Eurozone ETF	Eurozone Equity	FLAT
23	EFV	0.0168	iShares MSCI EAFE Value ETF	Developed ex-US Equity	FLAT
26	JPIN	0.0168	JPMorgan Diversified Return Internation...	International Equity	LONG
18	IQDF	0.0167	FlexShares International Quality Divide...	International Equity	FLAT
22	SCZ	0.0167	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	FLAT
24	FNDC	0.0167	Schwab Fundamental International Small ...	International Equity	LONG
37	RODM	0.0167	Hartford Multifactor Developed Markets ...	International Equity	LONG
38	VYMI	0.0167	Vanguard International High Dividend Yi...	International Dividend E...	LONG
44	SCHC	0.0167	Schwab International Small-Cap Equity E...	Developed ex-US Equity	FLAT
56	HFXI	0.0167	NYLI FTSE International Equity Currency...	International Equity	FLAT
20	EFG	0.0166	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	FLAT
34	IVLU	0.0166	iShares MSCI Intl Value Factor ETF	International Equity	FLAT
36	FEZ	0.0166	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	FLAT
45	VSS	0.0166	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	LONG
4	FEP	0.0165	First Trust Europe AlphaDEX Fund	Europe Equity	LONG
7	DLS	0.0165	WisdomTree International SmallCap Fund	International Equity	LONG
31	IMTM	0.0164	iShares MSCI Intl Momentum Factor ETF	International Equity	FLAT
43	VIGI	0.0163	Vanguard International Dividend Appreci...	International Dividend E...	FLAT
1	EWG	0.0162	iShares MSCI Germany Index Fund	Germany Equity	FLAT
12	EWQ	0.0162	iShares MSCI France Index Fund	France Equity	FLAT
21	GWX	0.0162	State Street SPDR S&P International Sma...	International Equity	LONG
5	VIDI	0.0160	Vident International Equity Strategy ETF	International Equity	FLAT
59	IPKW	0.0160	Invesco International BuyBack Achievers...	International Equity	FLAT
61	EWI	0.0159	iShares MSCI Italy ETF	Italy Equity	LONG
13	EWU	0.0158	iShares MSCI United Kingdom ETF	United Kingdom Equity	FLAT
15	IDV	0.0158	iShares International Select Dividend E...	International Dividend E...	FLAT
17	EUFN	0.0158	iShares MSCI Europe Financials ETF	Europe Financials	LONG
27	DBEF	0.0157	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
42	HDEF	0.0157	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	FLAT
58	EWN	0.0157	iShares MSCI Netherlands Index Fund	Netherlands Equity	LONG
10	HEZU	0.0156	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	FLAT
28	HEFA	0.0156	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT

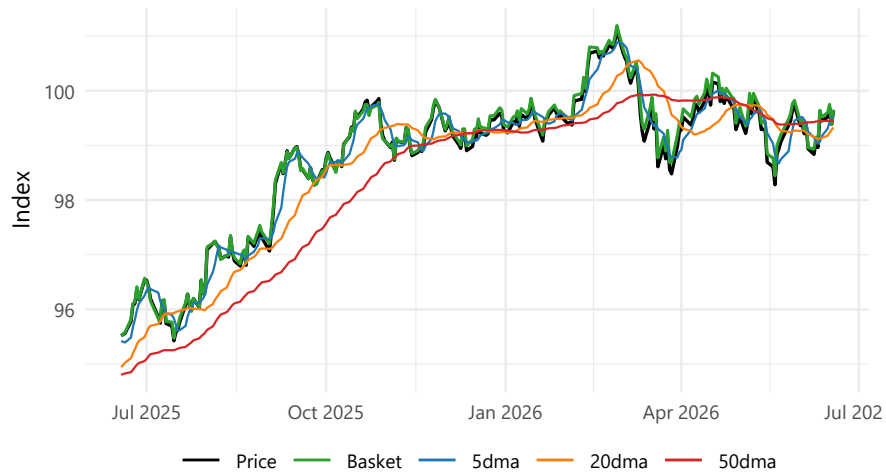
55	EWD	0.0156	iShares MSCI Sweden ETF	Sweden Equity	FLAT
8	HEDJ	0.0151	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
9	IHDG	0.0151	WisdomTree International Hedged Quality...	Alternatives	LONG
11	EWP	0.0150	iShares MSCI Spain ETF	Spain Equity	FLAT
57	EWO	0.0148	iShares MSCI Austria ETF	Austria Equity	LONG
60	EWK	0.0146	iShares MSCI Belgium ETF	Belgium Equity	FLAT
54	EPOL	0.0135	iShares MSCI Poland ETF	Poland Equity	FLAT

Cluster 4 – Aggregate Fixed Income

Members: 58 **Var explained:** 0.832 **Signal:** LONG since 2026-06-17 (1 days, 0.3%)

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +4.5%)
- Latest sell_cross: 2026-05-13
- Current 252d return: +4.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-17 > last sell at 2026-05-13)



Category breakdown

category	n_members	abs_weight
Fixed Income	19	0.3283
Investment Grade Corporate Bond	13	0.2302
Aggregate Fixed Income	6	0.1120
Mortgage-Backed Fixed Income	6	0.0982
Active Fixed Income	5	0.0891
Cash / Ultra-Short Fixed Income	3	0.0470
International Fixed Income	2	0.0313
Target Maturity Bond	2	0.0307
Agency Fixed Income	1	0.0166
Taxable Municipal Bond	1	0.0165

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.006

ticker	rep_wt	name	category	etf_signal
BND	1.007	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	BIV	0.0188	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
26	AGG	0.0188	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	LONG
28	BND	0.0188	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG
30	SCHZ	0.0188	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	LONG
20	IUSB	0.0187	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	LONG
31	SPAB	0.0187	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	LONG
2	IEF	0.0185	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
17	IGIB	0.0184	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	FLAT
24	VCIT	0.0184	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	FLAT
7	FBND	0.0183	Fidelity Total Bond ETF	Active Fixed Income	LONG
43	SPTI	0.0183	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
46	SCHR	0.0183	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
47	VGIT	0.0183	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
13	AGGY	0.0182	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	LONG
15	SPIB	0.0182	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	FLAT
6	GTO	0.0181	Invesco Total Return Bond ETF	Active Fixed Income	LONG
11	GVI	0.0181	iShares Intermediate Government/Credit ...	Fixed Income	FLAT
40	IEI	0.0181	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
50	MBB	0.0181	iShares MBS ETF	Mortgage-Backed Fixed In...	LONG
12	QLTA	0.0180	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	LONG

27	USIG	0.0180	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	LONG
51	VMBS	0.0180	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	LONG
52	SPMB	0.0180	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	LONG
3	LQD	0.0179	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	LONG
5	BOND	0.0179	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	LONG
19	ISTB	0.0178	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT
8	FLCO	0.0177	Franklin Investment Grade Corporate ETF	Active Fixed Income	LONG
10	CORP	0.0177	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	LONG
22	BLV	0.0176	Vanguard Long-Term Bond ETF	Fixed Income	LONG
41	TLH	0.0176	iShares 10-20 Year Treasury Bond ETF	Fixed Income	LONG
45	GOVT	0.0176	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
35	IGSB	0.0175	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
37	VCSH	0.0175	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
23	BSV	0.0174	Vanguard Short-Term Bond ETF	Fixed Income	FLAT
49	GNMA	0.0174	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	FLAT
18	IGLB	0.0172	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	LONG
16	SPLB	0.0171	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	LONG
25	VCLT	0.0171	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	LONG
36	SLQD	0.0171	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
44	SPTL	0.0171	State Street SPDR Portfolio Long Term T...	Fixed Income	LONG
48	VGLT	0.0171	Vanguard Long-Term Treasury ETF	Fixed Income	LONG
29	TOTL	0.0170	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
1	TLT	0.0169	iShares 20+ Year Treasury Bond ETF	Fixed Income	LONG
39	AGZ	0.0166	iShares Agency Bond ETF	Agency Fixed Income	FLAT
9	BAB	0.0165	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	LONG
42	EDV	0.0163	Vanguard Extended Duration Treasury ETF	Fixed Income	LONG
34	SPSB	0.0162	State Street SPDR Portfolio Short Term ...	Fixed Income	FLAT
53	SHY	0.0162	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
56	VGSH	0.0159	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
57	BNDX	0.0158	Vanguard Total International Bond ETF	International Fixed Inco...	LONG
33	BSCQ	0.0156	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	FLAT
55	SCHO	0.0156	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
58	IAGG	0.0156	iShares International Aggregate Bond Fu...	International Fixed Inco...	LONG
54	SPTS	0.0155	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
14	IBDR	0.0151	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	FLAT
38	CMBS	0.0136	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	FLAT
4	LMBS	0.0133	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	LONG
32	LDUR	0.0121	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	LONG

Cluster 5 – U.S. Large Cap Equity

Members: 57 Var explained: 0.905 Signal: FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +25.2%)
- Latest sell_cross: 2026-06-10
- Current 252d return: +22.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-10 more recent than last effective buy 2026-04-13)

**Category breakdown**

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1654
U.S. Total Market Equity	7	0.1289
Global Equity	7	0.1265
Consumer Discretionary	6	0.0968
Asset Allocation	4	0.0671
U.S. Mega Cap Equity	3	0.0542
U.S. Quality / Factor Equity	3	0.0535
U.S. Multifactor Equity	3	0.0535
Communication Services	3	0.0482
U.S. ESG Equity	2	0.0367
U.S. Equity	2	0.0366
Commodity	1	0.0184
Large Cap Value Equities	1	0.0182
Large Cap Blend Equities	1	0.0182
Dividend Equity	1	0.0178

U.S. Dividend Equity	1	0.0175
Alternatives	1	0.0158
International IPO Equity	1	0.0141
Singapore Equity	1	0.0128

Replication basket – VOO (cor=0.99) basket lead at lag 1: -0.010

ticker	rep_wt	name	category	etf_signal
VOO	0.958	Vanguard S&P 500 ETF	U.S. Large Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	SPY	0.0184	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	FLAT
30	IWV	0.0184	iShares Russell 3000 Fund	U.S. Total Market Equity	FLAT
31	IYY	0.0184	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	FLAT
36	ESGU	0.0184	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	FLAT
37	IWB	0.0184	iShares Russell 1000 ETF	U.S. Large Cap Equity	FLAT
42	VTHR	0.0184	Vanguard Russell 3000 ETF	U.S. Total Market Equity	FLAT
43	GSLC	0.0184	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	FLAT
45	VONE	0.0184	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	FLAT
47	VV	0.0184	Vanguard Large-Cap ETF	U.S. Large Cap Equity	FLAT
48	ILCB	0.0184	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	FLAT
49	ITOT	0.0184	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	FLAT
50	IVV	0.0184	iShares Core S&P 500 ETF	U.S. Large Cap Equity	FLAT
51	SCHB	0.0184	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	FLAT
52	SCHX	0.0184	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	FLAT
53	SPTM	0.0184	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	FLAT
54	VOO	0.0184	Vanguard S&P 500 ETF	U.S. Large Cap Equity	FLAT
55	VTI	0.0184	Vanguard Total Stock Market ETF	U.S. Total Market Equity	FLAT
12	URTH	0.0183	iShares MSCI World ETF	Global Equity	FLAT
26	JHML	0.0183	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	FLAT
29	SUSA	0.0183	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	FLAT
34	SPYX	0.0183	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	FLAT
38	IWL	0.0183	iShares Russell Top 200 ETF	U.S. Equity	FLAT
46	MGC	0.0183	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	FLAT
7	ACWI	0.0182	iShares MSCI ACWI ETF	Global Equity	FLAT
15	VT	0.0182	Vanguard Total World Stock Index ETF	Global Equity	FLAT
22	TILT	0.0182	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	FLAT
28	DSI	0.0182	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	FLAT
32	LRGF	0.0182	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	FLAT

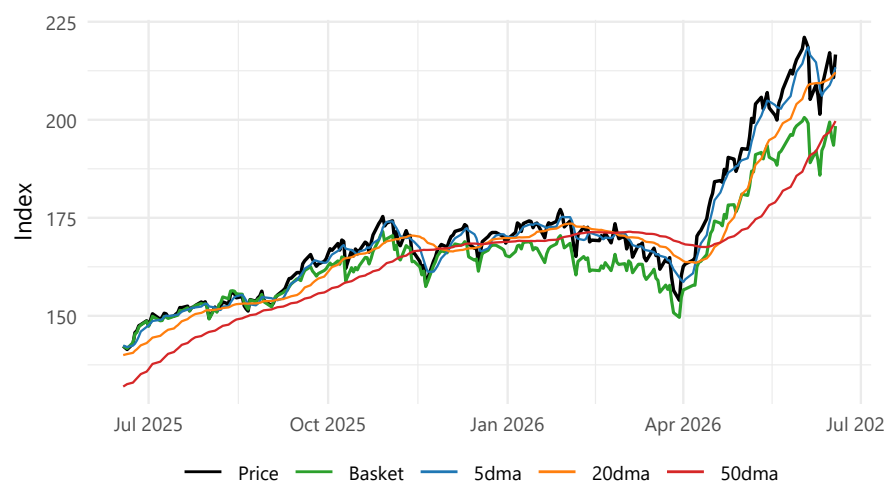
44	EPS	0.0182	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	FLAT
13	CRBN	0.0181	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	FLAT
14	SPGM	0.0181	State Street SPDR Portfolio MSCI Global...	Global Equity	FLAT
33	OEF	0.0181	iShares S&P 100 Fund	U.S. Mega Cap Equity	FLAT
39	QUAL	0.0181	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	LONG
8	AOA	0.0180	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	FLAT
23	IOO	0.0178	iShares Global 100 ETF	Global Equity	FLAT
24	QDF	0.0178	FlexShares Quality Dividend Index Fund	Dividend Equity	FLAT
25	CATH	0.0178	Global X S&P 500 Catholic Values ETF	Global Equity	FLAT
35	XLG	0.0178	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	FLAT
40	QUS	0.0178	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	FLAT
41	SPHQ	0.0176	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	LONG
11	AOR	0.0175	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	FLAT
27	DGRW	0.0175	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	FLAT
17	IYC	0.0171	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	FLAT
57	OMFL	0.0171	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	FLAT
18	VCR	0.0168	Vanguard Consumer Discretion ETF	Consumer Discretionary	LONG
21	FDIS	0.0168	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	LONG
1	XLY	0.0166	State Street Consumer Discretionary Sel...	Consumer Discretionary	FLAT
10	AOM	0.0164	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	FLAT
19	VOX	0.0162	Vanguard Communication Services ETF	Communication Services	FLAT
20	FCOM	0.0162	Fidelity MSCI Communication Services In...	Communication Services	FLAT
2	XLC	0.0158	State Street Communication Services Sel...	Communication Services	FLAT
5	QAI	0.0158	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	FLAT
56	RTH	0.0155	VanEck Retail ETF	Consumer Discretionary	FLAT
9	AOK	0.0151	iShares Core 30/70 Conservative Allocat...	Asset Allocation	FLAT
16	FPXI	0.0141	First Trust International Equity Opport...	International IPO Equity	FLAT
3	ESPO	0.0139	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
6	EWS	0.0128	iShares MSCI Singapore ETF	Singapore Equity	LONG

Cluster 6 – U.S. Large Cap Growth

Members: 39 **Var explained:** 0.896 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +55.3%)
- Latest sell_cross: 2026-06-11
- Current 252d return: +52.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-11 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Technology	19	0.4861
U.S. Large Cap Growth	11	0.2915
Growth Equity	2	0.0522
U.S. Equity	1	0.0267
U.S. Mega Cap Growth	1	0.0264
U.S. Equal Weight Equity	1	0.0257
Options Income	1	0.0239
U.S. Small Cap Technology	1	0.0236
Trend-Following Equity	1	0.0235
U.S. Trend-Following Equity	1	0.0204

Replication basket – QQQ (cor=0.98) basket lead at lag 1: -0.031

ticker	rep_wt	name	category	etf_signal
QQQ	1.022	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	QQQ	0.0268	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	FLAT
17	IGM	0.0268	iShares Expanded Tech Sector ETF	Technology	FLAT
25	VGT	0.0268	Vanguard Information Tech ETF	Technology	FLAT
27	FTEC	0.0268	Fidelity MSCI Information Technology In...	Technology	FLAT
19	IYW	0.0267	iShares U.S. Technology ETF	Technology	FLAT
21	ONEQ	0.0267	Fidelity Nasdaq Composite Index ETF	U.S. Equity	FLAT
1	XLK	0.0266	State Street Technology Select Sector S...	Technology	FLAT
18	IXN	0.0266	iShares Global Tech ETF	Technology	FLAT
30	ILCG	0.0266	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
23	IWF	0.0265	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	FLAT
24	IVW	0.0265	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	FLAT
26	VOOG	0.0265	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	FLAT
28	VONG	0.0265	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	FLAT
31	IUSG	0.0265	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	FLAT
33	SPYG	0.0265	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	FLAT
34	VUG	0.0265	Vanguard Growth ETF	U.S. Large Cap Growth	FLAT
13	XNTK	0.0264	State Street SPDR NYSE Technology ETF	Technology	LONG
20	NULG	0.0264	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
29	MGK	0.0264	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	FLAT
32	SCHG	0.0264	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
22	IWY	0.0263	iShares Russell Top 200 Growth ETF	Growth Equity	FLAT
8	QTEC	0.0262	First Trust NASDAQ-100-Technology Secto...	Technology	LONG
4	AIQ	0.0259	Global X Artificial Intelligence & Tech...	Technology	FLAT
12	FXL	0.0259	First Trust Technology AlphaDEX	Technology	LONG
16	PWB	0.0259	Invesco Large Cap Growth ETF	Growth Equity	FLAT
7	QQEW	0.0257	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	LONG
10	TDIV	0.0253	First Trust NASDAQ Technology Dividend ...	Technology	FLAT
3	SMH	0.0252	VanEck Semiconductor ETF	Technology	LONG
2	SOXX	0.0249	iShares PHLX SOX Semiconductor Sector I...	Technology	LONG
39	QTUM	0.0248	Defiance Quantum ETF	Technology	LONG
9	PSI	0.0244	Invesco Semiconductors ETF	Technology	LONG
11	XSD	0.0243	State Street SPDR S&P Semiconductor ETF	Technology	LONG
37	FTXL	0.0243	First Trust Nasdaq Semiconductor ETF	Technology	LONG
6	NXTG	0.0241	First Trust Indxx NextG ETF	Technology	LONG
38	PTF	0.0241	Invesco Dorsey Wright Technology Moment...	Technology	LONG
15	QYLD	0.0239	Global X NASDAQ 100 Covered Call ETF	Options Income	FLAT
36	PSCT	0.0236	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	LONG
14	PTNQ	0.0235	Pacer Trendpilot 100 ETF	Trend-Following Equity	FLAT
35	PTLC	0.0204	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	FLAT

Cluster 7 – U.S. Extended Market Equity**Members:** 28 **Var explained:** 0.812 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-04-14 – filter **PASS** (252d at date = +31.5%)
- Latest sell_cross: 2026-06-10
- Current 252d return: +20.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-10 more recent than last effective buy 2026-04-14)

**Category breakdown**

category	n_members	abs_weight
Technology	15	0.5270
U.S. Small Cap Growth	3	0.1131
Convertible Bond	2	0.0726
Financials	2	0.0722
U.S. Extended Market Equity	1	0.0382
Growth Equity	1	0.0378
U.S. Equity	1	0.0369
IPO Equity	1	0.0368
Consumer Discretionary	1	0.0361
Israel Equity	1	0.0292

Replication basket – VXF (cor=0.96) basket lead at lag 1: 0.020

ticker	rep_wt	name	category	etf_signal
VXF	1.07	Vanguard Extended Market ETF	U.S. Extended Market ...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

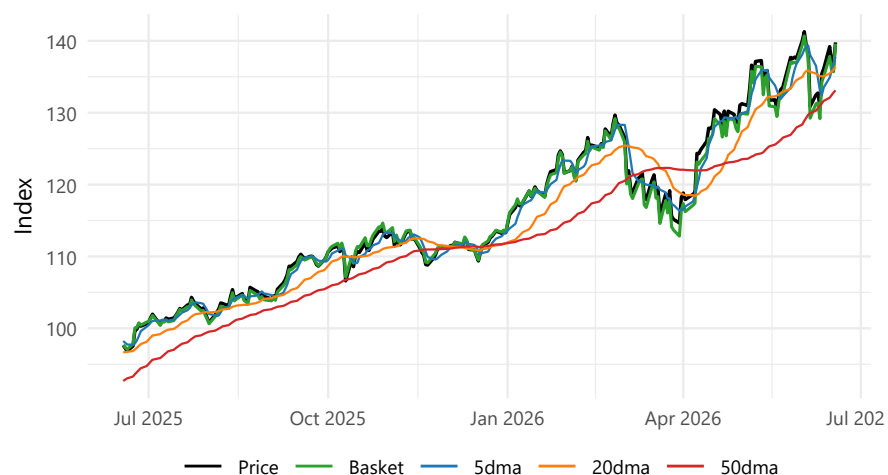
	ticker	PC_wt	name	category	etf_signal
23	VBK	0.0384	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	FLAT
25	VXF	0.0382	Vanguard Extended Market ETF	U.S. Extended Market Equ...	FLAT
24	ISCG	0.0378	iShares Morningstar Small-Cap Growth ETF	Growth Equity	FLAT
11	XT	0.0376	iShares Future Exponential Technologies...	Technology	FLAT
21	IWO	0.0374	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
22	VTWG	0.0373	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
18	FINX	0.0371	Global X FinTech ETF	Financials	FLAT
26	SPHB	0.0369	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG
14	IPO	0.0368	Renaissance IPO ETF	IPO Equity	LONG
19	CWB	0.0366	State Street SPDR Bloomberg Convertible...	Convertible Bond	FLAT
8	ARKW	0.0365	ARK Next Generation Internet ETF	Technology	LONG
16	FDN	0.0365	First Trust DJ Internet Index Fund	Technology	FLAT
7	ARKK	0.0364	ARK Innovation ETF	Technology	LONG
2	SKYY	0.0363	First Trust Cloud Computing ETF	Technology	FLAT
4	ROBO	0.0363	ROBO Global Robotics and Automation Ind...	Technology	FLAT
9	ARKQ	0.0363	ARK Autonomous Technology & Robotics ETF	Technology	LONG
15	PNQI	0.0362	Invesco Nasdaq Internet ETF	Technology	FLAT
12	IBUY	0.0361	Amplify Online Retail ETF	Consumer Discretionary	LONG
20	ICVT	0.0360	iShares Convertible Bond ETF	Convertible Bond	FLAT
3	BOTZ	0.0359	Global X Robotics & Artificial Intellig...	Technology	FLAT
17	IPAY	0.0351	Amplify Digital Payments ETF	Financials	FLAT
10	PRNT	0.0349	3D Printing (The) ETF	Technology	FLAT
1	IGV	0.0348	iShares Expanded Tech-Software Sector E...	Technology	FLAT
13	CIBR	0.0340	First Trust NASDAQ Cybersecurity ETF	Technology	FLAT
6	HACK	0.0337	Amplify Cybersecurity ETF	Technology	LONG
5	BLOK	0.0329	Amplify Blockchain Technology ETF	Technology	FLAT
27	EIS	0.0292	iShares MSCI Israel ETF	Israel Equity	FLAT
28	BLCN	0.0286	Siren NexGen Economy ETF	Technology	LONG

Cluster 8 – Emerging Markets Equity

Members: 25 **Var explained:** 0.873 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +57.0%)
- Latest sell_cross: 2026-06-10
- Current 252d return: +43.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-10 more recent than last effective buy 2026-04-10)

**Category breakdown**

category	n_members	abs_weight
Emerging Markets Equity	19	0.7726
Asia Equity	2	0.0813
Asia ex-Japan Equity	1	0.0418
Taiwan Equity	1	0.0360
Global Dividend Equity	1	0.0342
South Korea Equity	1	0.0341

Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.029

ticker	rep_wt	name	category	etf_signal
IEMG	0.924	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

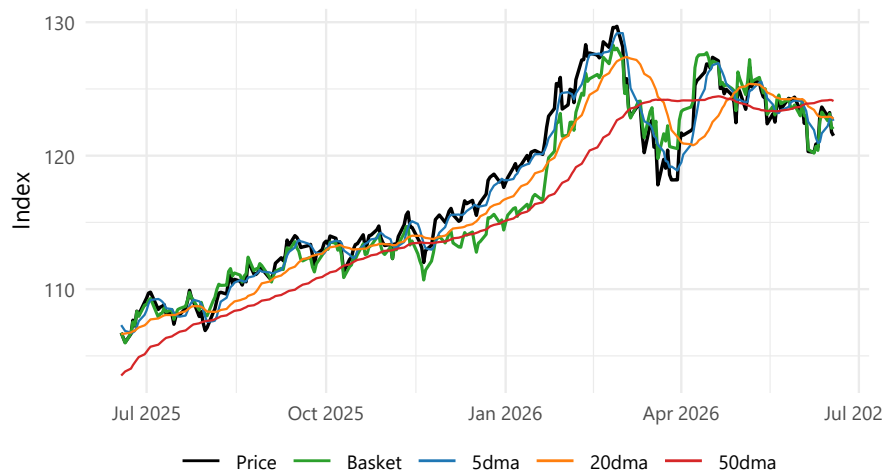
	ticker	PC_wt	name	category	etf_signal
21	IEMG	0.0425	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	FLAT
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	FLAT
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	LONG
16	GEM	0.0422	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	FLAT
20	ESGE	0.0422	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	FLAT
23	SPEM	0.0422	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	FLAT
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	LONG
18	XSOE	0.0420	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	FLAT
5	AAXJ	0.0418	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	FLAT
15	EMGF	0.0418	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	FLAT
11	EEMA	0.0414	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	FLAT
14	GMF	0.0411	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	FLAT
12	PXH	0.0410	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	LONG
17	FNDE	0.0410	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	LONG
19	EEMV	0.0404	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	FLAT
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	LONG
10	AIA	0.0402	iShares Asia 50 ETF	Asia Equity	FLAT
7	DEM	0.0397	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	LONG
6	EWX	0.0392	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	LONG
4	FEM	0.0387	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	FLAT
13	DVYE	0.0365	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	FLAT
9	EWT	0.0360	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
25	FEMS	0.0348	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	FLAT
24	SDIV	0.0342	Global X SuperDividend ETF	Global Dividend Equity	FLAT
1	EWY	0.0341	iShares MSCI South Korea ETF	South Korea Equity	FLAT

Cluster 9 – Asia-Pacific Equity

Members: 23 **Var explained:** 0.685 **Signal:** LONG since 2026-05-22 (18 days, -1.6%)

Signal logic:

- Latest buy_cross: 2026-05-22 – filter **PASS** (252d at date = +17.2%)
- Latest sell_cross: 2026-05-18
- Current 252d return: +13.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-22 > last sell at 2026-05-18)



Category breakdown

category	n_members	abs_weight
Asia-Pacific Equity	2	0.0978
International Dividend Equity	2	0.0959
International Equity	2	0.0956
Global Equity	2	0.0919
Real Estate	2	0.0917
Emerging Markets Fixed Income	2	0.0837
Currency	2	0.0813
Australia Equity	1	0.0485
Canada Equity	1	0.0462
Switzerland Equity	1	0.0431
South Africa Equity	1	0.0411
Latin America Equity	1	0.0396
New Zealand Equity	1	0.0395
Mexico Equity	1	0.0374
Brazil Equity	1	0.0335
Chile Equity	1	0.0332

Replication basket – EWZ+IGF+BBAX (cor=0.97) basket lead at lag 1: -0.004

ticker	rep_wt	name	category	etf_signal
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EWZ	0.095	iShares MSCI Brazil ETF	Brazil Equity	FLAT
IGF	0.289	iShares Global Infrastructure ETF	Global Equity	FLAT
BBAX	0.453	JPMorgan BetaBuilders Developed Asia ...	Asia-Pacific Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	EPP	0.0489	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	FLAT
23	BBAX	0.0489	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	FLAT
11	IDLV	0.0488	Invesco S&P International Developed Low...	International Equity	FLAT
4	EWA	0.0485	iShares MSCI Australia Index Fund	Australia Equity	FLAT
10	DWX	0.0482	State Street SPDR S&P International Div...	International Dividend E...	LONG
3	PID	0.0477	Invesco International Dividend Achiever...	International Dividend E...	FLAT
12	EFAV	0.0468	iShares MSCI EAFE Min Vol Factor ETF	International Equity	FLAT
8	NFRA	0.0466	FlexShares STOXX Global Broad Infrastru...	Global Equity	LONG
13	VNQI	0.0464	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	FLAT
6	EWC	0.0462	iShares MSCI Canada Index Fund	Canada Equity	LONG
9	IGF	0.0454	iShares Global Infrastructure ETF	Global Equity	FLAT
2	RWX	0.0453	State Street SPDR Dow Jones Internation...	Real Estate	FLAT
5	EWL	0.0431	iShares MSCI Switzerland ETF	Switzerland Equity	LONG
19	FXA	0.0431	Invesco CurrencyShares Australian Dolla...	Currency	FLAT
14	EBND	0.0420	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	LONG
15	EMLC	0.0417	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	LONG
21	EZA	0.0411	iShares MSCI South Africa Index Fund	South Africa Equity	FLAT
18	ILF	0.0396	iShares Latin America 40 ETF	Latin America Equity	FLAT
20	ENZL	0.0395	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
22	FXC	0.0381	Invesco CurrencyShares Canadian Dollar ...	Currency	FLAT
16	EWX	0.0374	iShares MSCI Mexico ETF	Mexico Equity	FLAT
1	EWZ	0.0335	iShares MSCI Brazil ETF	Brazil Equity	FLAT
17	ECH	0.0332	iShares MSCI Chile ETF	Chile Equity	FLAT

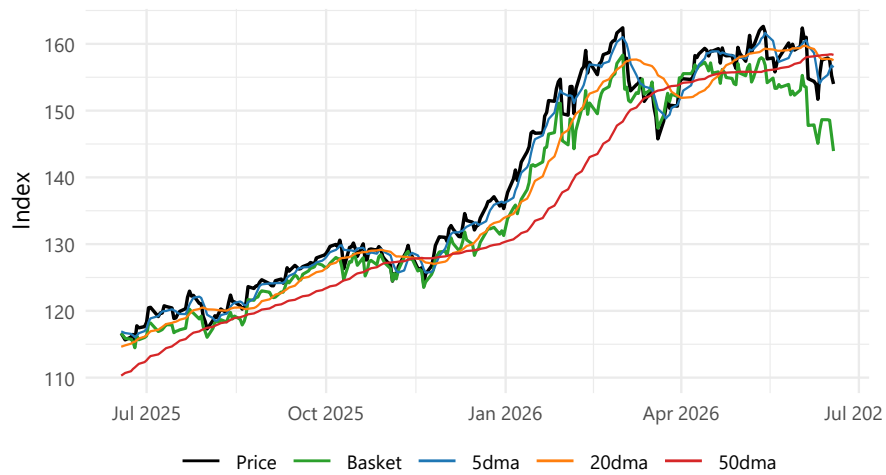
Cluster 10 – Materials

Members: 21 **Var explained:** 0.703 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +44.4%)
- Latest sell_cross: 2026-06-09
- Current 252d return: +31.8% – filter **ALLOWS** new entries today

- Decision: **FLAT** (last sell 2026-06-09 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Materials	13	0.6540
MLP / Energy Infrastructure	4	0.1691
Energy	1	0.0501
Consumer Staples	1	0.0493
Norway Equity	1	0.0439
Industrial Metals	1	0.0335

Replication basket – AMLP+URA+GUNR (cor=0.96) basket lead at lag 1: 0.043

ticker	rep_wt	name	category	etf_signal
AMLP	0.111	Alerian MLP ETF	MLP / Energy Infrastr...	FLAT
URA	0.083	Global X Uranium ETF	Materials	FLAT
GUNR	0.774	FlexShares Global Upstream Natural Re...	Materials	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

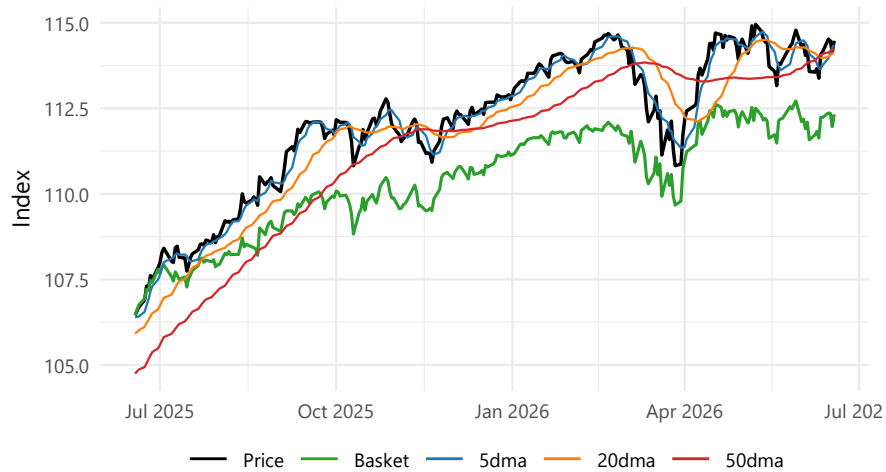
	ticker	PC_wt	name	category	etf_signal
15	GNR	0.0549	State Street SPDR S&P Global Natural Re...	Materials	FLAT
14	GUNR	0.0543	FlexShares Global Upstream Natural Reso...	Materials	FLAT
10	MXI	0.0528	iShares Global Materials ETF	Materials	LONG
11	IYM	0.0528	iShares U.S. Basic Materials ETF	Materials	LONG
21	NANR	0.0524	State Street SPDR S&P North American Na...	Materials	FLAT
13	VAW	0.0513	Vanguard Materials ETF	Materials	LONG
16	PICK	0.0509	iShares MSCI Global Select Metals & Min...	Materials	LONG
12	RSPM	0.0506	Invesco S&P 500 Equal Weight Materials ...	Materials	LONG
1	XLB	0.0504	State Street Materials Select Sector SP...	Materials	LONG
20	IGE	0.0501	iShares North American Natural Resource...	Energy	FLAT
4	XME	0.0499	State Street SPDR S&P Metals & Mining E...	Materials	FLAT
17	SLX	0.0497	VanEck Steel ETF	Materials	FLAT
6	MOO	0.0493	VanEck Agribusiness ETF	Consumer Staples	FLAT
3	COPX	0.0472	Global X Copper Miners ETF	Materials	LONG
18	NORW	0.0439	Global X MSCI Norway ETF	Norway Equity	FLAT
5	EMLP	0.0438	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	FLAT
9	MLPX	0.0437	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
7	AMLP	0.0414	Alerian MLP ETF	MLP / Energy Infrastruct...	FLAT
8	MLPA	0.0403	Global X MLP ETF	MLP / Energy Infrastruct...	LONG
2	URA	0.0367	Global X Uranium ETF	Materials	FLAT
19	DBB	0.0335	Invesco DB Base Metals Fund	Industrial Metals	FLAT

Cluster 11 – High Yield Bond

Members: 21 **Var explained:** 0.766 **Signal:** LONG since 2026-06-17 (1 days, 0.3%)

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +7.3%)
- Latest sell_cross: 2026-06-09
- Current 252d return: +7.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-17 > last sell at 2026-06-09)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.5155
Preferred Stock	7	0.3061
Emerging Markets Fixed Income	4	0.1784

Replication basket – HYG (cor=0.95) basket lead at lag 1: 0.048

ticker	rep_wt	name	category	etf_signal
HYG	0.916	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	JNK	0.0526	State Street SPDR Bloomberg High Yield ...	High Yield Bond	LONG
1	HYG	0.0525	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	LONG
19	HYLB	0.0524	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	LONG
18	BBHY	0.0523	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	LONG
16	SHYG	0.0521	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	LONG

14	SJNK	0.0520	State Street SPDR Bloomberg Short Term ...	High Yield Bond	LONG
15	ANGL	0.0516	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	LONG
17	FALN	0.0516	iShares Fallen Angels USD Bond ETF	High Yield Bond	LONG
6	HYS	0.0511	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	LONG
9	PFF	0.0480	iShares Preferred and Income Securities...	Preferred Stock	FLAT
3	HYLS	0.0473	First Trust Tactical High Yield ETF	High Yield Bond	LONG
2	EMB	0.0467	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
21	VWOB	0.0467	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	LONG
11	PFXF	0.0462	VanEck Preferred Securities ex Financia...	Preferred Stock	LONG
20	PCY	0.0457	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	LONG
7	PGX	0.0446	Invesco Preferred ETF	Preferred Stock	FLAT
10	PSK	0.0441	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0434	First Trust Preferred Securities and In...	Preferred Stock	LONG
5	PGF	0.0431	Invesco Financial Preferred ETF	Preferred Stock	FLAT
12	HYEM	0.0393	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	LONG
8	VRP	0.0368	Invesco Variable Rate Preferred ETF	Preferred Stock	FLAT

Cluster 12 – Healthcare

Members: 17 **Var explained:** 0.761 **Signal:** LONG since 2026-05-06 (30 days, 2.9%)

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.8%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +18.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-06 > last sell at 2026-04-29)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9419
U.S. Small Cap Healthcare	1	0.0581

Replication basket – VHT (cor=0.93) basket lead at lag 1: -0.006

ticker	rep_wt	name	category	etf_signal
VHT	1.099	Vanguard Health Care ETF	Healthcare	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	VHT	0.0654	Vanguard Health Care ETF	Healthcare	LONG
16	FHLC	0.0653	Fidelity MSCI Health Care Index ETF	Healthcare	LONG
9	FXH	0.0646	First Trust Health Care AlphaDEX	Healthcare	LONG
14	RSPH	0.0640	Invesco S&P 500 Equal Weight Health Car...	Healthcare	LONG
13	IYH	0.0638	iShares U.S. Healthcare ETF	Healthcare	LONG
10	IXJ	0.0623	iShares Global Healthcare ETF	Healthcare	LONG

1	XLV	0.0621	State Street Health Care Select Sector ...	Healthcare	LONG
3	IBB	0.0604	iShares Biotechnology ETF	Healthcare	LONG
6	FBT	0.0596	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0581	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	LONG
7	XHE	0.0574	State Street SPDR S&P Health Care Equip...	Healthcare	LONG
12	IHI	0.0571	iShares U.S. Medical Devices ETF	Healthcare	FLAT
17	PPH	0.0543	VanEck Pharmaceutical ETF	Healthcare	LONG
5	PTH	0.0542	Invesco Dorsey Wright Healthcare Moment...	Healthcare	LONG
4	XBI	0.0534	State Street SPDR S&P Biotech ETF	Healthcare	LONG
2	ARKG	0.0508	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0472	iShares U.S. Health Care Providers ETF	Healthcare	LONG

Cluster 13 – China Equity

Members: 15 **Var explained:** 0.802 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-15 – filter **PASS** (252d at date = +24.3%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +8.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-04-15)



Category breakdown

category	n_members	abs_weight
China Equity	13	0.8766
Emerging Markets Equity	1	0.0643
Hong Kong Equity	1	0.0591

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.011

ticker	rep_wt	name	category	etf_signal
KWEB	0.563	KraneShares CSI China Internet ETF	China Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	CXSE	0.0737	WisdomTree China ex-State-Owned Enterpr...	China Equity	FLAT
9	GXC	0.0735	State Street SPDR S&P China ETF	China Equity	FLAT
1	MCHI	0.0729	iShares MSCI China ETF	China Equity	FLAT
2	FXI	0.0711	iShares China Large-Cap ETF	China Equity	FLAT
5	CQQQ	0.0711	Invesco China Technology ETF	China Equity	FLAT
7	CHIQ	0.0706	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0680	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0669	Xtrackers Harvest CSI 300 China A-Share...	China Equity	LONG
8	CNYA	0.0661	iShares MSCI China A ETF	China Equity	LONG
13	KGRN	0.0651	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0643	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
10	KBA	0.0595	KraneShares Boseri MSCI China A 50 Conn...	China Equity	FLAT
14	CNXT	0.0594	VanEck ChiNext Innovators ETF	China Equity	FLAT
12	EWK	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	FLAT
15	KURE	0.0586	KraneShares MSCI All China Health Care ...	China Equity	FLAT

Cluster 14 – Financials

Members: 14 **Var explained:** 0.877 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +19.9%)
- Latest sell_cross: 2026-05-11
- Current 252d return: +15.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-11 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Financials	13	0.938
Global Equity	1	0.062

Replication basket – XLF (cor=0.96) basket lead at lag 1: 0.011

ticker	rep_wt	name	category	etf_signal
XLF	1.097	State Street Financial Select Sector ...	Financials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
10	VFH	0.0755	Vanguard Financials ETF	Financials	FLAT

11	FNCL	0.0754	Fidelity MSCI Financials Index ETF	Financials	LONG
7	IYF	0.0752	iShares U.S. Financial ETF	Financials	FLAT
5	FXO	0.0751	First Trust Financials AlphaDEX	Financials	LONG
8	IYG	0.0747	iShares U.S. Financial Services ETF	Financials	FLAT
1	XLF	0.0740	State Street Financial Select Sector SP...	Financials	LONG
9	KBWB	0.0733	Invesco KBW Bank ETF	Financials	FLAT
3	KBE	0.0712	State Street SPDR S&P Bank ETF	Financials	FLAT
14	KCE	0.0708	State Street SPDR S&P Capital Markets E...	Financials	FLAT
6	IAT	0.0705	iShares U.S. Regional Banks ETF	Financials	LONG
13	IAI	0.0698	iShares U.S. Broker-Dealers & Securitie...	Financials	LONG
2	KRE	0.0687	State Street SPDR S&P Regional Banking ...	Financials	FLAT
4	KIE	0.0638	State Street SPDR S&P Insurance ETF	Financials	LONG
12	PSP	0.0620	Invesco Global Listed Private Equity ETF	Global Equity	FLAT

Cluster 15 – Real Estate

Members: 14 **Var explained:** 0.962 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +11.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: -0.000

ticker	rep_wt	name	category	etf_signal
VNQ	0.971	Vanguard Real Estate ETF	Real Estate	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	FLAT
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	FLAT
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	FLAT
9	FREL	0.0724	Fidelity MSCI Real Estate Index ETF	Real Estate	FLAT
10	USRT	0.0724	iShares Core U.S. REIT ETF	Real Estate	FLAT
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	FLAT
13	PSR	0.0721	Invesco Active U.S. Real Estate Fund	Real Estate	FLAT

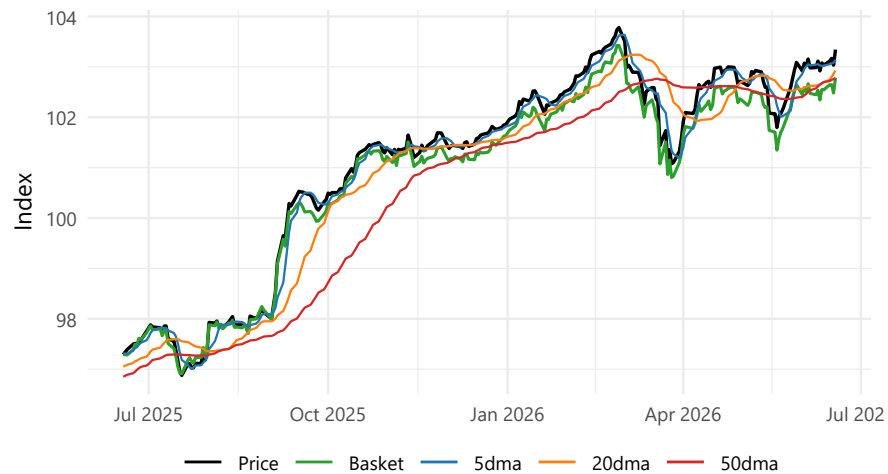
1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	FLAT
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	FLAT
7	REET	0.0715	iShares Global REIT ETF	Real Estate	FLAT
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	LONG
4	REZ	0.0698	iShares Residential and Multisector Rea...	Real Estate	FLAT
14	NURE	0.0688	Nuveen Short-Term REIT ETF	Real Estate	LONG
12	INDS	0.0683	Pacer Industrial Real Estate ETF	Real Estate	LONG

Cluster 16 – Municipal Bond

Members: 13 **Var explained:** 0.819 **Signal:** LONG since 2026-05-28 (15 days, 0.4%)

Signal logic:

- Latest buy_cross: 2026-05-28 – filter **PASS** (252d at date = +5.9%)
- Latest sell_cross: 2026-05-15
- Current 252d return: +6.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-28 > last sell at 2026-05-15)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.178

ticker	rep_wt	name	category	etf_signal
MUB	0.904	iShares National Muni Bond ETF	Municipal Bond	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

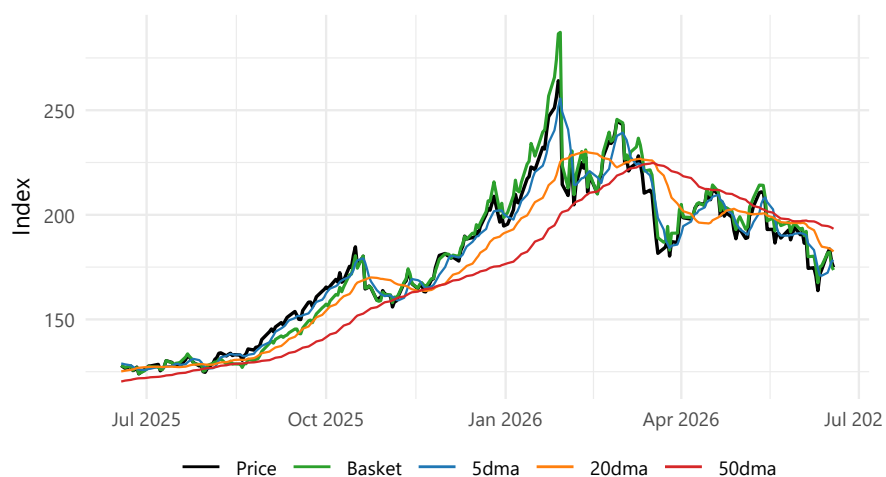
	ticker	PC_wt	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	LONG
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	LONG
6	CMF	0.0801	iShares California Muni Bond ETF	Municipal Bond	LONG
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	LONG
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	LONG
1	FMB	0.0795	First Trust Managed Municipal ETF	Municipal Bond	LONG
7	NYF	0.0793	iShares New York Muni Bond ETF	Municipal Bond	LONG
5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	LONG
4	MUNI	0.0769	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	LONG
2	HYD	0.0748	VanEck High Yield Muni ETF	Municipal Bond	LONG
3	HYMB	0.0720	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	LONG
12	SUB	0.0695	iShares Short-Term National Muni Bond E...	Municipal Bond	LONG
10	SHM	0.0667	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	LONG

Cluster 17 – Gold

Members: 12 **Var explained:** 0.861 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-12 – filter **PASS** (252d at date = +75.1%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +35.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-05-12)



Category breakdown

category	n_members	abs_weight
Materials	5	0.4158
Gold	3	0.2501
Precious Metals	2	0.1724
Silver	2	0.1616

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.014

ticker	rep_wt	name	category	etf_signal
GLD	0.006	SPDR Gold Shares	Gold	FLAT
IAU	0.767	iShares Gold Trust Shares	Gold	FLAT
SLV	0.412	iShares Silver Trust	Silver	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	DBP	0.0866	Invesco DB Precious Metals Fund	Precious Metals	FLAT
7	GLTR	0.0859	abrdn Physical Precious Metals Basket S...	Precious Metals	FLAT

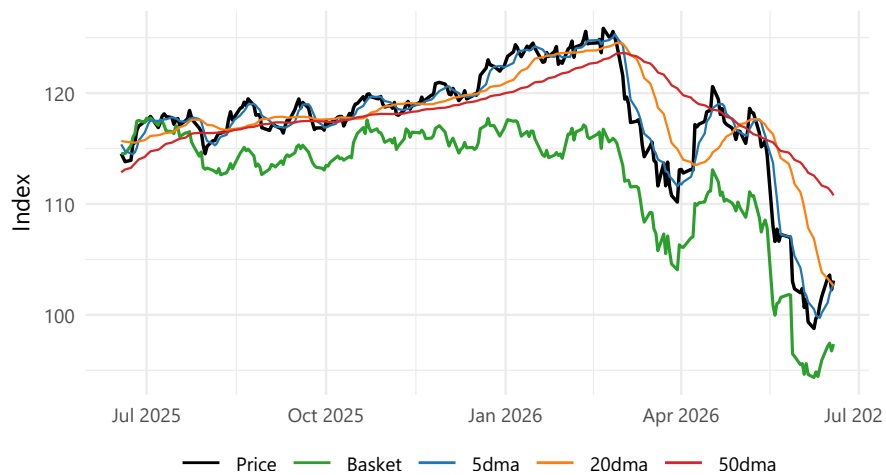
2	GDXJ	0.0841	VanEck Junior Gold Miners ETF	Materials	FLAT
1	GDX	0.0840	VanEck Gold Miners ETF	Materials	FLAT
3	GLD	0.0834	SPDR Gold Shares	Gold	FLAT
9	IAU	0.0834	iShares Gold Trust Shares	Gold	FLAT
10	SGOL	0.0833	abrdn Physical Gold Shares ETF	Gold	FLAT
12	GOAU	0.0832	US Global GO Gold and Precious Metal Mi...	Materials	FLAT
6	SIL	0.0828	Global X Silver Miners ETF	Materials	FLAT
5	SILJ	0.0818	Amplify Junior Silver Miners ETF	Materials	FLAT
4	SLV	0.0809	iShares Silver Trust	Silver	LONG
8	SIVR	0.0807	abrdn Physical Silver Shares ETF	Silver	LONG

Cluster 18 – India Equity

Members: 12 **Var explained:** 0.340 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +4.5%)
- Latest sell_cross: 2026-05-13
- Current 252d return: -10.1% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-05-13 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.4510
Malaysia Equity	1	0.1093
Indonesia Equity	1	0.1022
Philippines Equity	1	0.0984
Vietnam Equity	1	0.0730
Turkey Equity	1	0.0589
U.S. Dividend Equity	1	0.0532
Agricultural Commodities	1	0.0259
Platinum	1	0.0225
Natural Gas	1	0.0056

Replication basket – NOBL+INDA+PPLT (cor=0.91) basket lead at lag 1: -0.004

ticker	rep_wt	name	category	etf_signal
NOBL	0.078	ProShares S&P 500 Dividend Aristocrat...	U.S. Dividend Equity	FLAT
INDA	0.642	iShares MSCI India ETF	India Equity	FLAT
PPLT	0.028	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	INDA	0.1516	iShares MSCI India ETF	India Equity	FLAT
3	INDY	0.1500	iShares India 50 ETF	India Equity	FLAT
4	EPI	0.1494	WisdomTree India Earnings Fund	India Equity	FLAT
10	EWM	0.1093	iShares MSCI Malaysia Index Fund	Malaysia Equity	FLAT
8	EIDO	0.1022	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
9	EPHE	0.0984	iShares MSCI Philippines ETF	Philippines Equity	FLAT
12	VNM	0.0730	VanEck Vietnam ETF	Vietnam Equity	FLAT
11	TUR	0.0589	iShares MSCI Turkey ETF	Turkey Equity	FLAT
5	NOBL	0.0532	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
2	DBA	0.0259	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT
6	PPLT	0.0225	abrdn Physical Platinum Shares ETF	Platinum	FLAT
7	UNG	0.0056	United States Natural Gas Fund LP	Natural Gas	FLAT

Cluster 19 – Consumer Staples

Members: 11 **Var explained:** 0.744 **Signal:** LONG since 2026-06-15 (3 days, -1.5%)

Signal logic:

- Latest buy_cross: 2026-06-15 – filter **PASS** (252d at date = +7.1%)
- Latest sell_cross: 2026-05-29
- Current 252d return: +7.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-15 > last sell at 2026-05-29)

**Category breakdown**

category	n_members	abs_weight
Consumer Staples	6	0.5732
Utilities	4	0.3521
U.S. Small Cap Consumer Staples	1	0.0747

Replication basket – XLU+XLP (cor=0.99) basket lead at lag 1: -0.010

ticker	rep_wt	name	category	etf_signal
XLU	0.364	State Street Utilities Select Sector ...	Utilities	FLAT

XLP 0.588 State Street Consumer Staples Select ... Consumer Staples LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	VDC	0.0983	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0983	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0975	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0949	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG
3	KXI	0.0946	iShares Global Consumer Staples ETF	Consumer Staples	LONG
11	PBJ	0.0897	Invesco Food & Beverage ETF	Consumer Staples	FLAT
7	IDU	0.0890	iShares U.S. Utilities ETF	Utilities	FLAT
8	VPU	0.0878	Vanguard Utilities ETF	Utilities	FLAT
9	FUTY	0.0878	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
2	XLU	0.0874	State Street Utilities Select Sector SP...	Utilities	FLAT
10	PSCC	0.0747	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	LONG

Cluster 20 – Inflation-Protected Fixed Income

Members: 10 **Var explained:** 0.906 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-27 – filter **PASS** (252d at date = +4.8%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +3.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2026-05-27)



Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP+SCHP (cor=0.99) basket lead at lag 1: -0.015

ticker	rep_wt	name	category	etf_signal
VTIP	0.262	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT
SCHP	0.762	Schwab U.S. TIPS ETF	Inflation-Protected F...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	TDTF	0.1035	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	LONG
6	TIPX	0.1034	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	FLAT
3	TIP	0.1030	iShares TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
8	SCHP	0.1023	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	FLAT
5	TDTT	0.1008	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.0999	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	FLAT

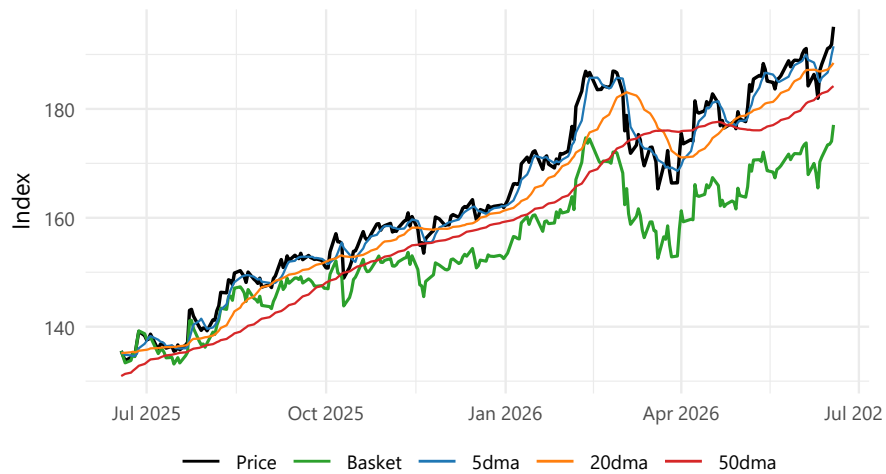
9	STIP	0.0998	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
2	STPZ	0.0996	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0994	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
1	LTPZ	0.0885	Pimco 15 Year U.S. TIPS Index Exchange-...	Inflation-Protected Fixe...	FLAT

Cluster 21 – Japan Equity

Members: 9 **Var explained:** 0.887 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +57.2%)
- Latest sell_cross: 2026-06-11
- Current 252d return: +45.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-11 more recent than last effective buy 2026-04-09)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7802

Asia-Pacific Equity	2	0.2198
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Replication basket – EWJ (cor=0.97) basket lead at lag 1: -0.040

ticker	rep_wt	name	category	etf_signal
EWJ	0.909	iShares MSCI Japan Index Fund	Japan Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

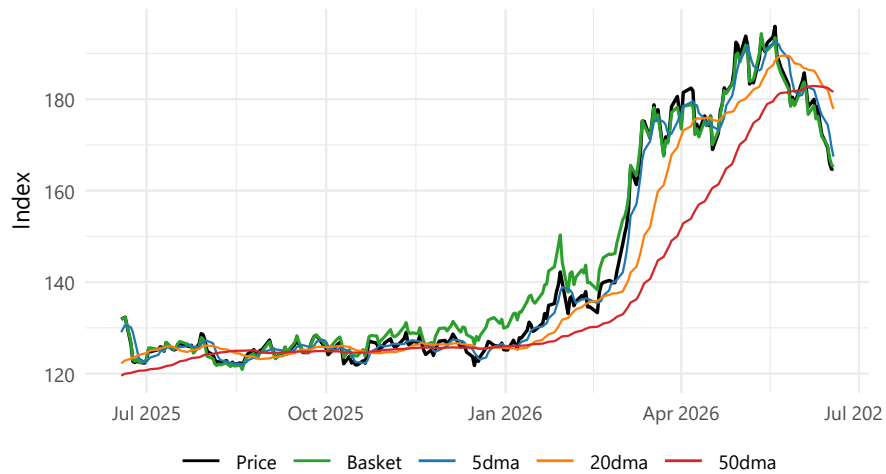
	ticker	PC_wt	name	category	etf_signal
1	EWJ	0.1142	iShares MSCI Japan Index Fund	Japan Equity	FLAT
8	BBJP	0.1141	JPMorgan BetaBuilders Japan ETF	Japan Equity	FLAT
9	HEWJ	0.1117	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
6	DBJP	0.1116	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
3	IPAC	0.1113	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	LONG
5	FLJH	0.1108	Franklin FTSE Japan Hedged ETF	Japan Equity	FLAT
2	DXJ	0.1097	WisdomTree Japan Hedged Equity Fund	Japan Equity	FLAT
4	VPL	0.1084	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	FLAT
7	EWJV	0.1081	iShares MSCI Japan Value ETF	Japan Equity	FLAT

Cluster 22 – Broad Commodities

Members: 9 **Var explained:** 0.912 **Signal:** LONG since 2026-06-04 (10 days, -9.7%)

Signal logic:

- Latest buy_cross: 2026-06-04 – filter **PASS** (252d at date = +51.6%)
- Latest sell_cross: 2026-05-27
- Current 252d return: +25.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-04 > last sell at 2026-05-27)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6698
Oil	2	0.2183
Energy Commodities	1	0.1119

Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.073

ticker	rep_wt	name	category	etf_signal
PDBC	1.176	Invesco Optimum Yield Diversified Com...	Broad Commodities	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	GSG	0.1151	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT
2	DBC	0.1150	Invesco DB Commodity Index Tracking Fund	Broad Commodities	LONG
7	PDBC	0.1149	Invesco Optimum Yield Diversified Commo...	Broad Commodities	LONG
8	COMT	0.1149	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
9	DBE	0.1119	Invesco DB Energy Fund	Energy Commodities	LONG

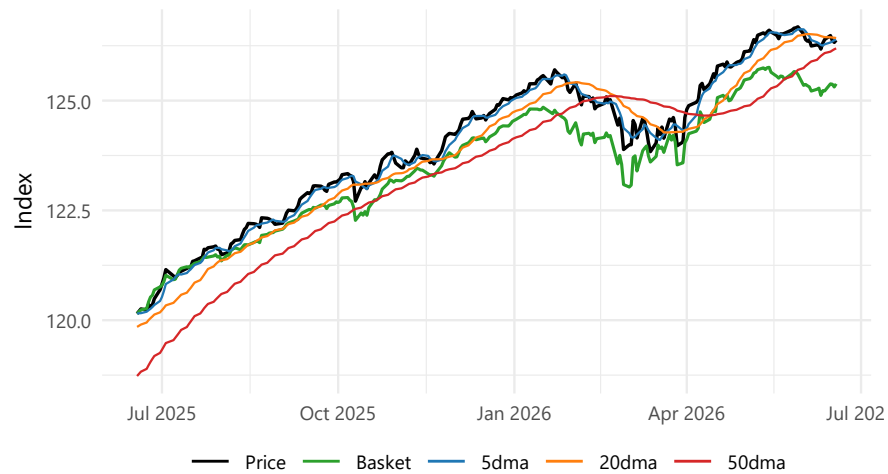
1	USO	0.1098	United States Oil Fund	Oil	LONG
4	DBO	0.1085	Invesco DB Oil Fund	Oil	FLAT
6	DJP	0.1052	iPath Bloomberg Commodity Index Total R...	Broad Commodities	LONG
3	FTGC	0.1047	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT

Cluster 23 – Bank Loan / Floating Rate

Members: 9 Var explained: 0.558 Signal: FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell_cross: 2026-06-04
- Current 252d return: +5.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-04 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3628
Interest Rate Hedged Fixed Income	3	0.3245

Floating Rate Fixed Income 3 0.3128

Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.047

ticker	rep_wt	name	category	etf_signal
FLOT	0.293	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	FLAT
BKLN	0.210	Invesco Senior Loan ETF	Bank Loan / Floating ...	LONG
SRLN	0.308	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	FLAT
FLRN	0.127	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	FLAT
FLTR	0.035	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

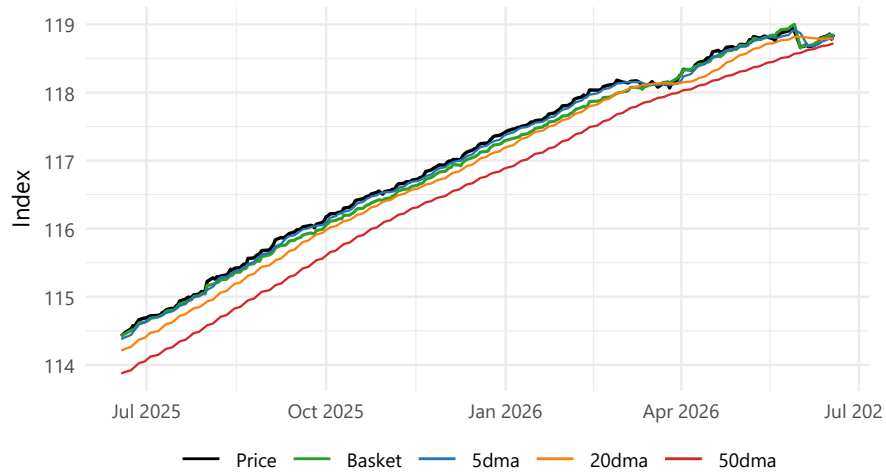
	ticker	PC_wt	name	category	etf_signal
2	SRLN	0.1250	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	FLAT
3	BKLN	0.1211	Invesco Senior Loan ETF	Bank Loan / Floating Rate	LONG
1	FTSL	0.1167	First Trust Senior Loan Fund	Bank Loan / Floating Rate	FLAT
5	LQDH	0.1145	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	FLAT
7	FLOT	0.1082	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	FLAT
6	IGBH	0.1057	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	FLAT
4	IGHG	0.1043	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	FLAT
8	FLRN	0.1037	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	FLAT
9	FLTR	0.1009	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 24 – Cash / Ultra-Short Fixed Income

Members: 9 **Var explained:** 0.454 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2022-10-28 – filter **BLOCKED** (252d at date = -0.5%; needs > 0)
- Latest effective buy: 2022-11-29 – delayed re-arm from blocked cross (252d at fire = +0.0%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +3.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2022-11-29)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.84) basket lead at lag 1: 0.092

ticker	rep_wt	name	category	etf_signal
SHV	0.297	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	FLAT
MINT	0.177	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	FLAT
GBIL	0.121	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	FLAT
ICSH	0.412	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	SHV	0.1281	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	FLAT
6	ICSH	0.1275	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	FLAT
5	GSY	0.1265	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	FLAT
3	FTSM	0.1204	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	FLAT

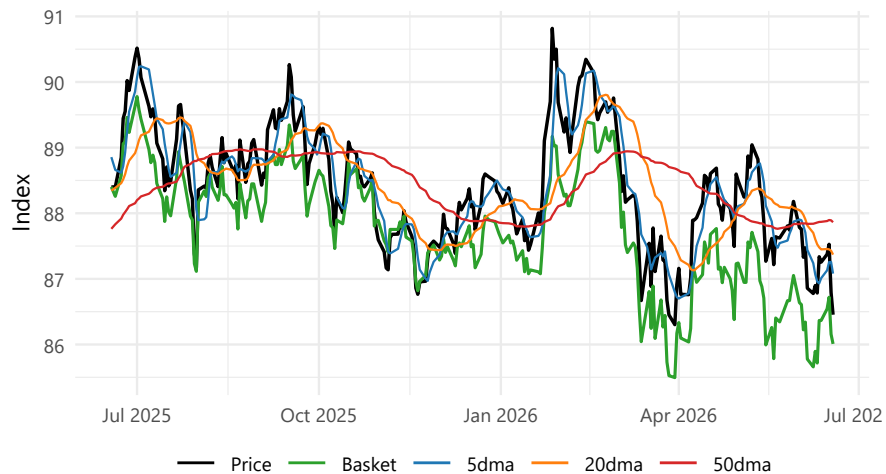
4	NEAR	0.1102	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT
2	MINT	0.1095	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	FLAT
1	VNLA	0.0992	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	FLAT
8	BIL	0.0945	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	FLAT
9	GBIL	0.0841	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	FLAT

Cluster 25 – International Fixed Income

Members: 6 **Var explained:** 0.742 **Signal:** LONG since 2026-05-29 (14 days, -2.0%)

Signal logic:

- Latest buy_cross: 2026-05-29 – filter **PASS** (252d at date = +0.6%)
- Latest sell_cross: 2026-05-18
- Current 252d return: -2.3% – filter **BLOCKS** new entries today
- Decision: **LONG** (effective buy at 2026-05-29 > last sell at 2026-05-18)



Category breakdown

category	n_members	abs_weight
Currency	4	0.6409

International Fixed Income	2	0.3591
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Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.026

ticker	rep_wt	name	category	etf_signal
BWX	0.751	State Street SPDR Bloomberg Internati...	International Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	FLAT
1	BWX	0.1790	State Street SPDR Bloomberg Internation...	International Fixed Inco...	FLAT
4	FXE	0.1702	Invesco CurrencyShares Euro Currency Tr...	Currency	FLAT
5	FXF	0.1649	Invesco CurrencyShares Swiss Franc Trust	Currency	FLAT
3	FXB	0.1611	Invesco CurrencyShares British Pound St...	Currency	FLAT
6	FXY	0.1448	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 26 – Leveraged / Inverse

Members: 2 **Var explained:** 0.625 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-02-05 – filter **BLOCKED** (252d at date = -32.0%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell_cross: 2026-06-18
- Current 252d return: -41.3% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-06-18 more recent than last effective buy 2025-04-08)



Category breakdown

category	n_members	abs_weight
Currency	1	0.5
Leveraged / Inverse	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.022

ticker	rep_wt	name	category	etf_signal
SQQQ	0.506	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.495	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
XLF	cluster_14	Financials	2026-04-13	47	3.7%	0.00	1.00
FNCL	cluster_14	Financials	2026-04-13	47	3.3%	0.00	1.00
FXO	cluster_14	Financials	2026-06-05	9	2.2%	0.00	1.00
IAT	cluster_14	Financials	2026-05-20	20	4.4%	0.00	1.00
IAI	cluster_14	Financials	2026-04-10	48	9.5%	0.00	1.00
KIE	cluster_14	Financials	2026-06-10	6	0.2%	0.01	0.96
XHB	cluster_1	U.S. Mid Cap Equity	2026-05-29	14	8.2%	0.00	1.00
PEJ	cluster_1	U.S. Mid Cap Equity	2026-05-26	17	6.2%	0.00	1.00
ITB	cluster_1	U.S. Mid Cap Equity	2026-06-09	7	3.1%	0.00	1.00
PSCD	cluster_1	U.S. Mid Cap Equity	2026-05-27	16	6.0%	0.00	1.00
PDP	cluster_1	U.S. Mid Cap Equity	2026-04-08	50	16.6%	0.00	1.00
XRT	cluster_1	U.S. Mid Cap Equity	2026-05-28	15	0.9%	0.00	1.00
ITA	cluster_1	U.S. Mid Cap Equity	2026-05-26	17	4.1%	0.00	1.00
SCHA	cluster_1	U.S. Mid Cap Equity	2026-04-09	49	14.2%	0.00	1.00
RSPD	cluster_1	U.S. Mid Cap Equity	2026-06-11	5	0.6%	0.00	1.00
SCHM	cluster_1	U.S. Mid Cap Equity	2026-04-09	49	11.1%	0.00	1.00
SDVY	cluster_1	U.S. Mid Cap Equity	2026-06-08	8	2.0%	0.00	1.00
VO	cluster_1	U.S. Mid Cap Equity	2026-04-13	47	6.8%	0.00	1.00
VFMO	cluster_1	U.S. Mid Cap Equity	2026-04-08	50	15.4%	0.00	1.00
IJS	cluster_1	U.S. Mid Cap Equity	2026-04-10	48	8.5%	0.00	1.00
FXR	cluster_1	U.S. Mid Cap Equity	2026-06-12	4	1.0%	0.00	1.00
IMCB	cluster_1	U.S. Mid Cap Equity	2026-04-10	48	9.9%	0.00	1.00
MTUM	cluster_1	U.S. Mid Cap Equity	2026-04-08	50	30.4%	0.00	1.00
FXD	cluster_1	U.S. Mid Cap Equity	2026-05-27	16	1.8%	0.00	1.00
PKB	cluster_1	U.S. Mid Cap Equity	2026-06-16	2	1.7%	0.00	0.99
IWR	cluster_1	U.S. Mid Cap Equity	2026-04-10	48	7.8%	0.01	0.99
ISMD	cluster_1	U.S. Mid Cap Equity	2026-04-09	49	14.4%	0.01	0.99
FTC	cluster_1	U.S. Mid Cap Equity	2026-04-10	48	17.1%	0.01	0.99
RPG	cluster_1	U.S. Mid Cap Equity	2026-04-09	49	23.0%	0.01	0.99
IMCG	cluster_1	U.S. Mid Cap Equity	2026-04-09	49	16.6%	0.01	0.99
MDYG	cluster_1	U.S. Mid Cap Equity	2026-04-09	49	9.7%	0.02	0.98
VOT	cluster_1	U.S. Mid Cap Equity	2026-04-14	46	10.6%	0.03	0.97
IJK	cluster_1	U.S. Mid Cap Equity	2026-04-09	49	9.5%	0.03	0.97
IVOG	cluster_1	U.S. Mid Cap Equity	2026-04-09	49	9.8%	0.03	0.97
XMMO	cluster_1	U.S. Mid Cap Equity	2026-04-06	52	16.5%	0.06	0.94
PPA	cluster_1	U.S. Mid Cap Equity	2026-05-27	16	0.5%	0.07	0.93
XAR	cluster_1	U.S. Mid Cap Equity	2026-05-22	18	4.8%	0.12	0.88

SIZE	cluster_1	U.S. Mid Cap Equity	2026-04-13	47	5.6%	0.12	0.88
FTXR	cluster_1	U.S. Mid Cap Equity	2026-04-17	43	6.9%	0.16	0.84
IWP	cluster_1	U.S. Mid Cap Equity	2026-04-15	45	4.9%	0.26	0.73
FV	cluster_1	U.S. Mid Cap Equity	2026-04-14	46	13.6%	0.53	0.47
PSCH	cluster_12	Healthcare	2026-04-13	47	13.2%	0.00	1.00
IHF	cluster_12	Healthcare	2026-05-08	28	4.6%	0.01	0.99
RSPH	cluster_12	Healthcare	2026-05-21	19	2.3%	0.02	0.98
XBI	cluster_12	Healthcare	2026-06-16	2	4.0%	0.00	0.97
IBB	cluster_12	Healthcare	2026-06-17	1	0.1%	0.00	0.96
FXH	cluster_12	Healthcare	2026-04-10	48	3.2%	0.04	0.96
VHT	cluster_12	Healthcare	2026-05-19	21	2.3%	0.07	0.93
FHLC	cluster_12	Healthcare	2026-05-19	21	2.0%	0.09	0.91
XHE	cluster_12	Healthcare	2026-06-09	7	-0.9%	0.08	0.85
PTH	cluster_12	Healthcare	2026-06-17	1	0.4%	0.01	0.83
XLV	cluster_12	Healthcare	2026-05-21	19	0.8%	0.18	0.82
IYH	cluster_12	Healthcare	2026-05-20	20	1.3%	0.30	0.70
PPH	cluster_12	Healthcare	2026-05-21	19	-1.2%	0.46	0.52
IXJ	cluster_12	Healthcare	2026-05-22	18	-2.0%	0.58	0.32
MGV	cluster_2	U.S. Large Cap Value	2026-04-13	47	8.4%	0.00	1.00
VTV	cluster_2	U.S. Large Cap Value	2026-04-13	47	6.9%	0.00	1.00
SCHV	cluster_2	U.S. Large Cap Value	2026-04-10	48	9.7%	0.00	1.00
PWV	cluster_2	U.S. Large Cap Value	2026-04-02	53	7.5%	0.00	1.00
PKW	cluster_2	U.S. Large Cap Value	2026-05-20	20	1.5%	0.00	1.00
VONV	cluster_2	U.S. Large Cap Value	2026-04-10	48	9.0%	0.00	1.00
SDY	cluster_2	U.S. Large Cap Value	2026-05-20	20	1.6%	0.00	1.00
FIW	cluster_2	U.S. Large Cap Value	2026-06-16	2	-0.7%	0.00	1.00
IWD	cluster_2	U.S. Large Cap Value	2026-04-10	48	9.2%	0.00	1.00
AIVL	cluster_2	U.S. Large Cap Value	2026-04-10	48	6.9%	0.00	1.00
IWX	cluster_2	U.S. Large Cap Value	2026-04-10	48	9.9%	0.00	1.00
RWL	cluster_2	U.S. Large Cap Value	2026-04-10	48	7.6%	0.00	1.00
FEX	cluster_2	U.S. Large Cap Value	2026-04-09	49	8.8%	0.00	1.00
DIA	cluster_2	U.S. Large Cap Value	2026-04-14	46	6.2%	0.00	1.00
RSP	cluster_2	U.S. Large Cap Value	2026-04-14	46	5.2%	0.00	1.00
RPV	cluster_2	U.S. Large Cap Value	2026-04-20	42	2.6%	0.01	0.99
REGL	cluster_2	U.S. Large Cap Value	2026-06-12	4	-0.6%	0.00	0.99
ONEY	cluster_2	U.S. Large Cap Value	2026-04-10	48	4.5%	0.01	0.99
LVHD	cluster_2	U.S. Large Cap Value	2026-06-10	6	-1.3%	0.01	0.98
SPYD	cluster_2	U.S. Large Cap Value	2026-04-21	41	2.8%	0.03	0.97
SPHD	cluster_2	U.S. Large Cap Value	2026-06-08	8	0.1%	0.03	0.97
DTD	cluster_2	U.S. Large Cap Value	2026-04-13	47	4.5%	0.03	0.97
SDOG	cluster_2	U.S. Large Cap Value	2026-04-07	51	4.0%	0.03	0.97
FTA	cluster_2	U.S. Large Cap Value	2026-05-21	19	0.9%	0.04	0.96
PRF	cluster_2	U.S. Large Cap Value	2026-04-09	49	8.9%	0.04	0.96
FVD	cluster_2	U.S. Large Cap Value	2026-06-10	6	-0.7%	0.02	0.96
FNDX	cluster_2	U.S. Large Cap Value	2026-04-09	49	7.8%	0.05	0.95

JVAL	cluster_2	U.S. Large Cap Value	2026-04-10	48	14.8%	0.06	0.94
VIG	cluster_2	U.S. Large Cap Value	2026-04-14	46	5.1%	0.06	0.94
DGRO	cluster_2	U.S. Large Cap Value	2026-04-13	47	3.5%	0.07	0.93
DLN	cluster_2	U.S. Large Cap Value	2026-04-13	47	4.7%	0.09	0.91
VOOV	cluster_2	U.S. Large Cap Value	2026-04-14	46	4.5%	0.14	0.86
SPYV	cluster_2	U.S. Large Cap Value	2026-04-14	46	4.4%	0.15	0.85
SPLV	cluster_2	U.S. Large Cap Value	2026-06-11	5	-1.0%	0.04	0.82
LGLV	cluster_2	U.S. Large Cap Value	2026-06-12	4	-1.9%	0.02	0.80
VYM	cluster_2	U.S. Large Cap Value	2026-04-10	48	4.0%	0.21	0.79
RDIV	cluster_2	U.S. Large Cap Value	2026-04-21	41	2.9%	0.22	0.78
PFM	cluster_2	U.S. Large Cap Value	2026-04-14	46	4.2%	0.24	0.76
FTCS	cluster_2	U.S. Large Cap Value	2026-06-10	6	-0.7%	0.16	0.75
FDVV	cluster_2	U.S. Large Cap Value	2026-04-14	46	4.8%	0.27	0.73
VLUE	cluster_2	U.S. Large Cap Value	2026-04-09	49	30.8%	0.30	0.70
IUSV	cluster_2	U.S. Large Cap Value	2026-04-14	46	4.0%	0.30	0.70
XMLV	cluster_2	U.S. Large Cap Value	2026-06-12	4	-2.0%	0.05	0.63
IVE	cluster_2	U.S. Large Cap Value	2026-04-14	46	4.1%	0.43	0.57
ILCV	cluster_2	U.S. Large Cap Value	2026-04-13	47	5.1%	0.46	0.54
OUSA	cluster_2	U.S. Large Cap Value	2026-04-20	42	-0.1%	0.77	0.17
SMMV	cluster_2	U.S. Large Cap Value	2026-06-15	3	-1.3%	0.13	0.16
SCHD	cluster_2	U.S. Large Cap Value	2026-04-21	41	2.8%	0.84	0.15
FDL	cluster_2	U.S. Large Cap Value	2026-06-05	9	-2.5%	0.89	-0.33
SHM	cluster_16	Municipal Bond	2026-05-28	15	0.1%	0.00	1.00
CMF	cluster_16	Municipal Bond	2026-05-28	15	0.4%	0.01	0.99
HYMB	cluster_16	Municipal Bond	2026-05-26	17	1.0%	0.01	0.98
FMB	cluster_16	Municipal Bond	2026-05-29	14	0.5%	0.01	0.98
PZA	cluster_16	Municipal Bond	2026-05-29	14	0.9%	0.02	0.98
SUB	cluster_16	Municipal Bond	2026-05-27	16	0.1%	0.02	0.98
NYF	cluster_16	Municipal Bond	2026-05-28	15	0.5%	0.02	0.98
HYD	cluster_16	Municipal Bond	2026-05-22	18	1.4%	0.03	0.97
MUNI	cluster_16	Municipal Bond	2026-05-28	15	0.1%	0.02	0.96
MUB	cluster_16	Municipal Bond	2026-05-28	15	0.3%	0.03	0.96
TFI	cluster_16	Municipal Bond	2026-05-28	15	0.3%	0.04	0.95
VTEB	cluster_16	Municipal Bond	2026-05-28	15	0.3%	0.04	0.95
ITM	cluster_16	Municipal Bond	2026-05-28	15	0.2%	0.09	0.88
HEWJ	cluster_21	Japan Equity	2026-04-08	50	10.8%	0.00	1.00
DBJP	cluster_21	Japan Equity	2026-04-08	50	9.4%	0.00	1.00
IPAC	cluster_21	Japan Equity	2026-06-12	4	1.7%	0.17	0.78
EDV	cluster_4	Aggregate Fixed Income	2026-06-12	4	2.1%	0.00	1.00
IAGG	cluster_4	Aggregate Fixed Income	2026-05-22	18	1.0%	0.00	1.00
TLT	cluster_4	Aggregate Fixed Income	2026-06-16	2	0.6%	0.00	1.00
SPTL	cluster_4	Aggregate Fixed Income	2026-06-16	2	0.5%	0.00	1.00
VGLT	cluster_4	Aggregate Fixed Income	2026-06-16	2	0.5%	0.00	0.99
BLV	cluster_4	Aggregate Fixed Income	2026-06-15	3	0.7%	0.01	0.99
BAB	cluster_4	Aggregate Fixed Income	2026-06-16	2	0.1%	0.00	0.99

BNDX	cluster_4	Aggregate Fixed Income	2026-05-26	17	0.2%	0.02	0.98
TLH	cluster_4	Aggregate Fixed Income	2026-06-16	2	0.3%	0.00	0.98
AGGY	cluster_4	Aggregate Fixed Income	2026-06-16	2	-0.1%	0.00	0.96
IGLB	cluster_4	Aggregate Fixed Income	2026-06-12	4	0.3%	0.03	0.95
SPLB	cluster_4	Aggregate Fixed Income	2026-06-12	4	0.3%	0.04	0.95
VCLT	cluster_4	Aggregate Fixed Income	2026-06-12	4	0.3%	0.05	0.93
GTO	cluster_4	Aggregate Fixed Income	2026-06-16	2	-0.2%	0.01	0.92
SPMB	cluster_4	Aggregate Fixed Income	2026-06-16	2	-0.2%	0.02	0.85
LQD	cluster_4	Aggregate Fixed Income	2026-06-16	2	-0.0%	0.06	0.82
FBND	cluster_4	Aggregate Fixed Income	2026-06-17	1	0.3%	0.01	0.82
BOND	cluster_4	Aggregate Fixed Income	2026-06-17	1	0.3%	0.02	0.80
IUSB	cluster_4	Aggregate Fixed Income	2026-06-17	1	0.3%	0.03	0.80
FLCO	cluster_4	Aggregate Fixed Income	2026-06-16	2	-0.1%	0.07	0.80
AGG	cluster_4	Aggregate Fixed Income	2026-06-17	1	0.3%	0.04	0.75
BND	cluster_4	Aggregate Fixed Income	2026-06-17	1	0.3%	0.04	0.75
USIG	cluster_4	Aggregate Fixed Income	2026-06-16	2	-0.1%	0.08	0.75
LMBS	cluster_4	Aggregate Fixed Income	2026-06-16	2	-0.2%	0.00	0.75
SPAB	cluster_4	Aggregate Fixed Income	2026-06-18	0	0.0%	0.04	0.74
MBB	cluster_4	Aggregate Fixed Income	2026-06-17	1	0.3%	0.05	0.74
SCHZ	cluster_4	Aggregate Fixed Income	2026-06-18	0	0.0%	0.04	0.72
QLTA	cluster_4	Aggregate Fixed Income	2026-06-17	1	0.2%	0.08	0.72
VMBS	cluster_4	Aggregate Fixed Income	2026-06-17	1	0.3%	0.07	0.67
CORP	cluster_4	Aggregate Fixed Income	2026-06-16	2	-0.1%	0.10	0.65
LDUR	cluster_4	Aggregate Fixed Income	2026-06-16	2	-0.2%	0.17	0.29
EWT	cluster_8	Emerging Markets Equity	2026-04-08	50	44.0%	0.08	0.92
DEM	cluster_8	Emerging Markets Equity	2026-04-09	49	8.9%	0.10	0.90
DGS	cluster_8	Emerging Markets Equity	2026-06-15	3	-0.2%	0.08	0.86
SCHE	cluster_8	Emerging Markets Equity	2026-06-15	3	-0.3%	0.08	0.82
VWO	cluster_8	Emerging Markets Equity	2026-06-15	3	-0.1%	0.09	0.81
EWX	cluster_8	Emerging Markets Equity	2026-06-15	3	0.8%	0.15	0.72
PXH	cluster_8	Emerging Markets Equity	2026-06-15	3	-1.2%	0.29	0.42
FNDE	cluster_8	Emerging Markets Equity	2026-06-15	3	-1.5%	0.30	0.35
NURE	cluster_15	Real Estate	2026-04-13	47	10.4%	0.01	0.99
INDS	cluster_15	Real Estate	2026-06-08	8	1.0%	0.11	0.75
RWO	cluster_15	Real Estate	2026-06-09	7	-1.0%	0.09	0.73
EWO	cluster_3	Developed ex-US Equity	2026-04-10	48	11.9%	0.00	1.00
IHDG	cluster_3	Developed ex-US Equity	2026-05-06	30	4.6%	0.00	1.00
EWN	cluster_3	Developed ex-US Equity	2026-04-10	48	13.5%	0.00	1.00
EWI	cluster_3	Developed ex-US Equity	2026-04-08	50	7.0%	0.00	1.00
GSIE	cluster_3	Developed ex-US Equity	2026-06-12	4	0.5%	0.01	0.97
VYMI	cluster_3	Developed ex-US Equity	2026-06-12	4	-1.6%	0.07	0.88
DWM	cluster_3	Developed ex-US Equity	2026-06-15	3	0.1%	0.06	0.87
EUFN	cluster_3	Developed ex-US Equity	2026-06-12	4	-0.3%	0.08	0.86
JPIN	cluster_3	Developed ex-US Equity	2026-06-16	2	-1.1%	0.17	0.61
IQLT	cluster_3	Developed ex-US Equity	2026-06-12	4	-1.2%	0.17	0.60

TLTD	cluster_3	Developed ex-US Equity	2026-06-12	4	-1.5%	0.35	0.41
FNDC	cluster_3	Developed ex-US Equity	2026-06-16	2	-1.1%	0.20	0.40
INTF	cluster_3	Developed ex-US Equity	2026-06-12	4	-1.4%	0.49	0.08
FEP	cluster_3	Developed ex-US Equity	2026-04-09	49	0.7%	0.55	0.06
RODM	cluster_3	Developed ex-US Equity	2026-06-12	4	-1.1%	0.46	0.03
VSS	cluster_3	Developed ex-US Equity	2026-06-17	1	-0.2%	0.45	-0.12
DLS	cluster_3	Developed ex-US Equity	2026-06-17	1	-0.3%	0.42	-0.13
GWX	cluster_3	Developed ex-US Equity	2026-06-17	1	0.1%	0.47	-0.15
PSCC	cluster_19	Consumer Staples	2026-06-10	6	2.2%	0.00	1.00
IYK	cluster_19	Consumer Staples	2026-06-09	7	-1.3%	0.17	0.78
XLP	cluster_19	Consumer Staples	2026-06-10	6	-2.6%	0.19	0.63
VDC	cluster_19	Consumer Staples	2026-06-11	5	-2.0%	0.14	0.43
FSTA	cluster_19	Consumer Staples	2026-06-11	5	-2.5%	0.16	0.39
KXI	cluster_19	Consumer Staples	2026-06-11	5	-3.2%	0.57	-0.32
PCY	cluster_11	High Yield Bond	2026-05-28	15	1.6%	0.00	1.00
VWOB	cluster_11	High Yield Bond	2026-05-26	17	1.2%	0.00	1.00
HYEM	cluster_11	High Yield Bond	2026-04-13	47	1.4%	0.03	0.95
FALN	cluster_11	High Yield Bond	2026-06-12	4	0.4%	0.01	0.95
ANGL	cluster_11	High Yield Bond	2026-06-15	3	0.0%	0.02	0.84
JNK	cluster_11	High Yield Bond	2026-06-16	2	-0.0%	0.16	0.47
HYS	cluster_11	High Yield Bond	2026-06-16	2	0.0%	0.18	0.44
BBHY	cluster_11	High Yield Bond	2026-06-16	2	-0.1%	0.18	0.38
HYG	cluster_11	High Yield Bond	2026-06-17	1	0.4%	0.19	0.35
HYLS	cluster_11	High Yield Bond	2026-06-16	2	-0.2%	0.02	0.34
HYLB	cluster_11	High Yield Bond	2026-06-17	1	0.2%	0.19	0.28
SJNK	cluster_11	High Yield Bond	2026-06-17	1	0.2%	0.22	0.21
SHYG	cluster_11	High Yield Bond	2026-06-17	1	0.3%	0.23	0.21
FPE	cluster_11	High Yield Bond	2026-05-27	16	0.1%	0.05	0.02
PFXF	cluster_11	High Yield Bond	2026-06-16	2	-0.2%	0.66	-0.28
PSI	cluster_6	U.S. Large Cap Growth	2026-04-07	51	79.1%	0.00	1.00
SOXX	cluster_6	U.S. Large Cap Growth	2026-04-08	50	72.6%	0.01	0.99
FTXL	cluster_6	U.S. Large Cap Growth	2026-04-07	51	84.8%	0.02	0.98
SMH	cluster_6	U.S. Large Cap Growth	2026-04-08	50	56.0%	0.02	0.98
PTF	cluster_6	U.S. Large Cap Growth	2026-04-07	51	47.4%	0.08	0.92
QTEC	cluster_6	U.S. Large Cap Growth	2026-04-10	48	48.1%	0.10	0.90
QTUM	cluster_6	U.S. Large Cap Growth	2026-04-10	48	43.9%	0.12	0.88
XNTK	cluster_6	U.S. Large Cap Growth	2026-04-10	48	39.5%	0.15	0.85
PSCT	cluster_6	U.S. Large Cap Growth	2026-04-07	51	40.0%	0.24	0.76
FXL	cluster_6	U.S. Large Cap Growth	2026-04-09	49	27.8%	0.35	0.65
QQEW	cluster_6	U.S. Large Cap Growth	2026-04-16	44	16.1%	0.37	0.62
XSD	cluster_6	U.S. Large Cap Growth	2026-04-09	49	74.7%	0.57	0.43
NXTG	cluster_6	U.S. Large Cap Growth	2026-04-08	50	31.5%	0.63	0.37
SPHQ	cluster_5	U.S. Large Cap Equity	2026-04-10	48	13.5%	0.00	1.00
QUAL	cluster_5	U.S. Large Cap Equity	2026-04-14	46	6.2%	0.12	0.88
VCR	cluster_5	U.S. Large Cap Equity	2026-06-15	3	-0.8%	0.32	0.26

FDIS	cluster_5	U.S. Large Cap Equity	2026-06-15	3	-1.1%	0.33	0.25
EWS	cluster_5	U.S. Large Cap Equity	2026-06-17	1	0.7%	0.55	0.15
IBUY	cluster_7	U.S. Extended Market Eq...	2026-06-15	3	0.5%	0.01	0.98
SPHB	cluster_7	U.S. Extended Market Eq...	2026-04-10	48	26.8%	0.03	0.97
IPO	cluster_7	U.S. Extended Market Eq...	2026-04-13	47	28.9%	0.03	0.97
VTWG	cluster_7	U.S. Extended Market Eq...	2026-04-13	47	15.0%	0.04	0.96
IWO	cluster_7	U.S. Extended Market Eq...	2026-04-13	47	14.8%	0.05	0.95
BLCN	cluster_7	U.S. Extended Market Eq...	2026-04-15	45	14.0%	0.34	0.64
ARKK	cluster_7	U.S. Extended Market Eq...	2026-06-17	1	2.2%	0.20	0.60
HACK	cluster_7	U.S. Extended Market Eq...	2026-04-17	43	22.2%	0.88	0.12
ARKW	cluster_7	U.S. Extended Market Eq...	2026-06-15	3	-0.5%	0.57	0.10
ARKQ	cluster_7	U.S. Extended Market Eq...	2026-06-17	1	1.3%	0.92	-0.58
EMLC	cluster_9	Asia-Pacific Equity	2026-06-16	2	-0.9%	0.00	0.90
EBND	cluster_9	Asia-Pacific Equity	2026-06-17	1	-0.2%	0.00	0.84
DWX	cluster_9	Asia-Pacific Equity	2026-06-12	4	-1.6%	0.19	0.59
ENZL	cluster_9	Asia-Pacific Equity	2026-05-22	18	0.8%	0.44	0.38
EWC	cluster_9	Asia-Pacific Equity	2026-04-09	49	2.5%	0.42	0.37
NFRA	cluster_9	Asia-Pacific Equity	2026-06-15	3	-2.5%	0.71	-0.30
EWL	cluster_9	Asia-Pacific Equity	2026-06-11	5	-2.5%	0.76	-0.36
BKLN	cluster_23	Bank Loan / Floating Ra...	2026-06-01	13	-0.2%	0.59	-0.59
XLB	cluster_10	Materials	2026-06-16	2	-1.7%	0.00	0.98
RSPM	cluster_10	Materials	2026-06-16	2	-1.6%	0.00	0.97
IYM	cluster_10	Materials	2026-06-15	3	-2.0%	0.01	0.96
VAW	cluster_10	Materials	2026-06-16	2	-1.5%	0.00	0.94
COPX	cluster_10	Materials	2026-06-16	2	-4.8%	0.26	0.58
MXI	cluster_10	Materials	2026-06-17	1	-1.1%	0.22	0.39
PICK	cluster_10	Materials	2026-06-15	3	-4.4%	0.50	0.15
MLPX	cluster_10	Materials	2026-06-08	8	-1.5%	0.98	-0.95
MLPA	cluster_10	Materials	2026-06-08	8	-4.0%	1.00	-1.00
UUP	cluster_26	Leveraged / Inverse	2026-05-18	22	2.2%	0.00	1.00
SIVR	cluster_17	Gold	2026-05-27	16	-11.8%	0.99	-0.99
SLV	cluster_17	Gold	2026-05-27	16	-11.8%	0.99	-0.99
TDTF	cluster_20	Inflation-Protected Fix...	2026-05-28	15	-1.8%	0.97	-0.97
ASHR	cluster_13	China Equity	2026-06-15	3	0.6%	0.12	0.74
CNYA	cluster_13	China Equity	2026-06-15	3	-0.2%	0.10	0.72
DBE	cluster_22	Broad Commodities	2026-06-03	11	-14.4%	1.00	-1.00
USO	cluster_22	Broad Commodities	2026-06-03	11	-18.5%	1.00	-1.00
PDBC	cluster_22	Broad Commodities	2026-06-03	11	-8.6%	1.00	-1.00
DBC	cluster_22	Broad Commodities	2026-06-03	11	-8.8%	1.00	-1.00
DJP	cluster_22	Broad Commodities	2026-06-03	11	-8.4%	1.00	-1.00

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
KBWB	cluster_14	Financials	1.00	1.00
PSCH	cluster_12	Healthcare	1.00	1.00
RDVY	cluster_2	U.S. Large Cap Value	1.00	1.00
PSCC	cluster_19	Consumer Staples	1.00	1.00
SLYG	cluster_1	U.S. Mid Cap Equity	1.00	1.00
EWO	cluster_3	Developed ex-US Equity	1.00	1.00
SPHQ	cluster_5	U.S. Large Cap Equity	1.00	1.00
EDV	cluster_4	Aggregate Fixed Income	1.00	1.00
SHM	cluster_16	Municipal Bond	1.00	1.00
GSY	cluster_24	Cash / Ultra-Short Fixed ...	1.00	1.00
PCY	cluster_11	High Yield Bond	1.00	1.00
PSI	cluster_6	U.S. Large Cap Growth	1.00	1.00
HEWJ	cluster_21	Japan Equity	1.00	1.00
NURE	cluster_15	Real Estate	1.00	0.99
VXF	cluster_7	U.S. Extended Market Equi...	1.00	0.99
XLB	cluster_10	Materials	0.98	0.98
GMF	cluster_8	Emerging Markets Equity	1.00	0.95
EMLC	cluster_9	Asia-Pacific Equity	0.91	0.90
CNXT	cluster_13	China Equity	0.99	0.80
LTPZ	cluster_20	Inflation-Protected Fixed...	0.64	0.48

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
FXC	cluster_9	Asia-Pacific Equity	1.00	-1.00
NORW	cluster_10	Materials	1.00	-1.00
GSG	cluster_22	Broad Commodities	1.00	-1.00
IDV	cluster_3	Developed ex-US Equity	1.00	-1.00
KGRN	cluster_13	China Equity	1.00	-1.00
XLC	cluster_5	U.S. Large Cap Equity	1.00	-1.00

FXV	cluster_25	International Fixed Income	1.00	-1.00
TDTT	cluster_20	Inflation-Protected Fixed...	1.00	-1.00
PNQI	cluster_7	U.S. Extended Market Equi...	1.00	-1.00
EWM	cluster_18	India Equity	1.00	-1.00
DBP	cluster_17	Gold	1.00	-1.00
SDIV	cluster_8	Emerging Markets Equity	0.99	-0.99
ACWV	cluster_2	U.S. Large Cap Value	1.00	-0.98
SRLN	cluster_23	Bank Loan / Floating Rate	0.94	-0.94
PBJ	cluster_19	Consumer Staples	0.76	-0.76
IWY	cluster_6	U.S. Large Cap Growth	0.99	-0.75
PSP	cluster_14	Financials	0.75	-0.74
SQQQ	cluster_26	Leveraged / Inverse	0.67	-0.67
IHI	cluster_12	Healthcare	0.60	-0.54
VGSH	cluster_4	Aggregate Fixed Income	0.56	-0.53

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_18	12	FLAT	0.305	0.095	0.00021	
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_18	12	FLAT	0.487	0.077	0.00017	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.716	0.041	0.00007	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_25	6	FLAT	0.730	0.094	0.00022	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.616	0.219	0.00036	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_18	12	FLAT	0.596	0.172	0.00039	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.756	0.135	0.00032	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.661	0.187	0.00046	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.756	0.163	0.00040	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.707	0.193	0.00045	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.851	0.025	0.00000	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.851	0.084	0.00021	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.755	0.182	0.00045	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_18	12	FLAT	0.500	0.220	0.00056	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_4	58	LONG	0.846	0.129	0.00036	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_4	58	FLAT	0.876	0.161	0.00031	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_25	6	FLAT	0.819	0.204	0.00049	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_4	58	LONG	0.837	0.239	0.00047	
NOBL	ProShares S&P 500 Dividend Aris...	U.S. Dividend Equity	cluster_18	12	FLAT	0.445	0.275	0.00075	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	13	LONG	0.878	0.250	0.00045	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_18	12	FLAT	0.444	0.330	0.00083	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	13	LONG	0.878	0.264	0.00053	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_13	15	FLAT	0.802	0.272	0.00071	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_10	21	FLAT	0.746	0.361	0.00085	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_18	12	FLAT	0.508	0.360	0.00103	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_13	15	FLAT	0.856	0.259	0.00072	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_23	9	FLAT	0.865	0.327	0.00071	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_18	12	FLAT	0.567	0.412	0.00107	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_13	15	FLAT	0.904	0.242	0.00064	
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected F...	cluster_20	10	FLAT	0.918	0.228	0.00057	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20